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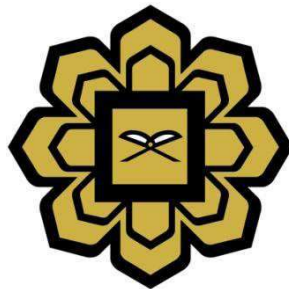
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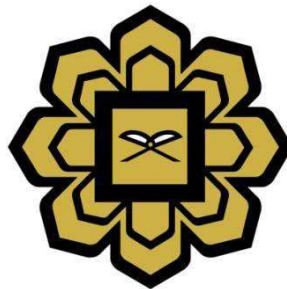


**DEPARTMENT OF MECHANICAL ENGINEERING
KULLIYAH OF ENGINEERING
INTERNATION ISLAMIC UNIVERSITY MALAYSIA
JANUARY 2013**

INVESTIGATION OF ENERGY ABSORPTION CAPABILITIES OF CRUSHABLE FOAM

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**A REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR A DEGREE OF BACHELOR OF
ENGINEERING (MECHANICAL - AUTOMOTIVE) (HONOURS)**

ABSTRACT

Crushable foams are suitable solution in the field of shock mitigation and impact absorption applications because of their incombustibility, cost, complex compression behavior and high energy absorption capabilities. Crushable foams are attracting increasing attention from engineers and researchers. In safety applications, accurate predictions of behavior of shock absorber materials are extremely important. In this report, the microstructure of the crushable foam material is discussed in order to understand its behavior under loading. This report focuses on the expanded polystyrene (EPS) crushable foams due to its wide range of applications and availability. A quasi-static compression test was conducted on the EPS foams with different compression rates and the resulting stress strain curves were compared to the ones found in the literature. The response of the EPS foams for different strain rates was analyzed. A gravity-driven drop test was carried out. LS-DYNA simulations were performed which were successfully validated by the experimental results. The model was discussed in details and was improved to match the experimental results to the simulation results. The model was extrapolated to simulate a car crash test in which the car runs into a bulk of crushable foam. The results were compared the results of a simulation of a car crashing into a rigid wall.

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CHAPTER 1

INTRODUCTION

1.1 Background

Crushable foams are extensively used in industry and in domestic applications as impact absorber materials because they have a high energy absorption capacity and complex compression behavior. However, they are not strong under tensile and shear loading and are rarely subjected intentionally to these type of loadings. Foam materials are made of two important variables: the matrix material and the morphology of the gaseous phase. There are two general morphologies of foam structure: open cell or closed cell. These cells can be considered as the basic units of crushable foam materials [1,2].

The matrix consists of interconnected solid struts which can be made of a variety of materials, rigid or flexible. In crushable foams, the matrix is fabricated by rigid materials. Therefore, they inherit the deformation behavior of these rigid materials and undergo deformation by ductile plasticity or brittle means, with little or no recovery. In the other hand, using flexible material –commonly plastic and rubber– as the matrix of the foam results in elastomeric foams which are highly flexible and have high level of recovery.

The mechanical behavior of crushable foams depends on the cell wall thickness, the size distribution and the cells shape, and the intrinsic properties of the polymer in the cell wall. Moreover, the density of the foam or the pore size affects the impact behavior, as it controls the rate at which the gas exits the foam during the compression. To use the

crushable foams widely in different applications, these parameters must be carefully tuned to achieve the desired behavior [3,4].

The impact event of the crushable foams is exemplified by a relatively large contact forces acting on a small contact area over a very short period of time. As a consequence of the insufficient understanding of structural response of foam materials under impact damage, the applications on crushable foams are limited in some cases. Impact damage is common. For an example, dropping a tool during manufacturing can be considered as low velocity impact damage. Thus, a detailed characterization of crushable foam deformation behavior is required to be able to use them efficiently in structural applications [5].

The objective of this research is to understand the structural response of low-density crushable foam materials under impact damage and to create a robust finite element model for crushable foam which can be used to simulate any experimental work.

To fully comprehend the compression behavior of crushable foams, the microstructure and the components of foam materials must be discussed in details. The expanded polystyrene (EPS) foam was studied in details due to its common use as an energy absorption material in safety application. The compression behavior and properties of EPS foam were found by conducting a simple quasi-static compression test. Also, the results were compared to the values available in the literature. A number of gravity-driven impact tests were carried out to further study the impact behavior of the crushable foams. And numerical simulations were carried to validate the results of the experiments and to study different cases of impacts.

CHAPTER 2

LITERATURE REVIEW

2.1 Mechanical Properties Crushable Foam

Foams are used in a wide range of various applications such as absorption energy, shock mitigation, protection of component, packaging, and in comfort situations. Thus, engineers and researchers have designed a wide variety of foams with widely varying properties designed especially to suit different application. The most common crushable foam is the polystyrene foam which by itself has a lot of grades and types. Crushable Polystyrene foams are utilized in protective applications. They are used often in packaging of components and products and in helmet liners. Even though they are easily damaged, they have high energy absorption capacity. The expanded polystyrene foams can be totally rigid or have a small elastic component, depend on the selection of the matrix material [3]. The mechanical properties of the expanded polystyrene crushable foam vary as the density changes. For expanded polystyrene crushable foam with the density of 19.8 kg/m³, the modulus of elasticity is 4.32 MPa while the Poisson's ratio is zero [6]. Since the Poisson's ratio is zero, with the assumption that the material is homogenous and isotropic, the other two elastic constant, shear modulus G, and bulk modulus K can be found easily using the following relationship [7]:

$$E = 2G(1 + \nu) = 3K(1 - 2\nu) \quad (1)$$

The following table illustrates these mechanical properties of crushable expanded polystyrene foam.

Table 1: The properties of expanded polystyrene foam

Properties	EPS foam
Density (kg/m ³)	19.8
Young's Modulus (MPa)	4.32
Poisson's Ratio	0.00
Shear Modulus (MPa)	2.16
Bulk Modulus (MPa)	1.44
Tensile Cut off stress (MPa)	0.10

The compression behavior of the all crushable foams, including the expanded polystyrene crushable foam, is strain rate dependant. Therefore, the stress-strain curve of any crushable foam material varies as the strain rate changes. This is illustrated by Figure 1 which shows different stress-strain curves of identical specimens of expanded polystyrene foam under compression. Each curve in the figure stands for a different strain rate. It is easy to notice that as the strain rate increases, the material becomes stiffer. Crushable expanded polystyrene foam with the density of 16 kg/m³ was used in the compression test to obtain the data in figure.

Of course, as the compression curve changes with the strain rate, other mechanical properties change as well. Figure 2 shows that yield strength of the expanded polystyrene crushable foam increases as the strain rate increases. The yield strength was obtained by examining the stress-strain curves. Further inspection of data in the figure shows that the yield stress increases linearly with the logarithm of the strain rate [3].

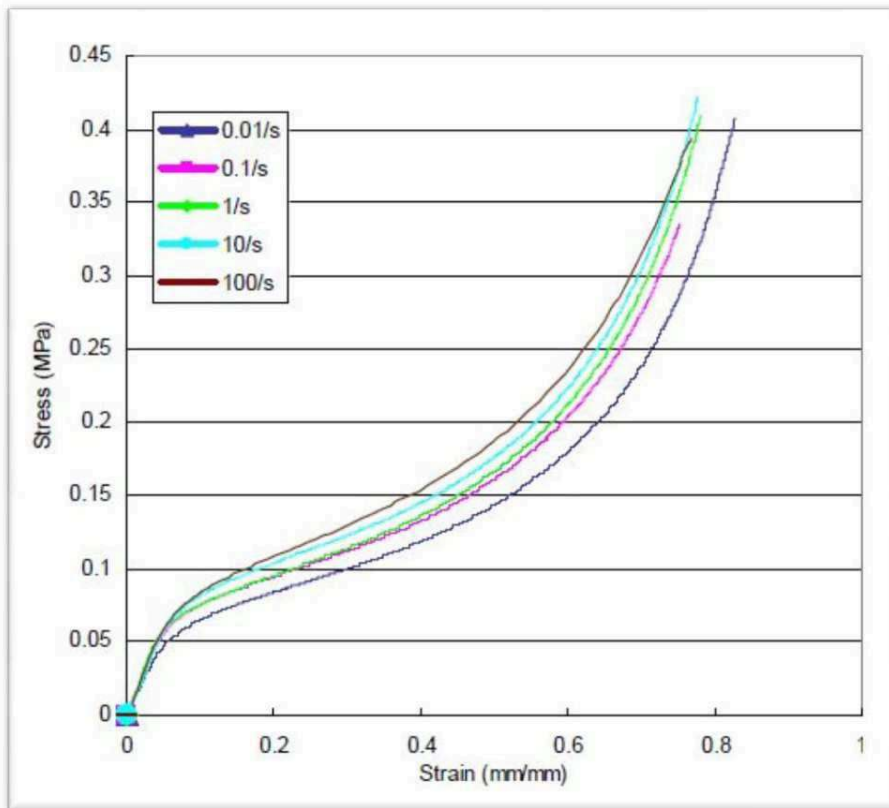


Figure 1: Rate dependency of the stress-strain curve of a crushable EPS foam.

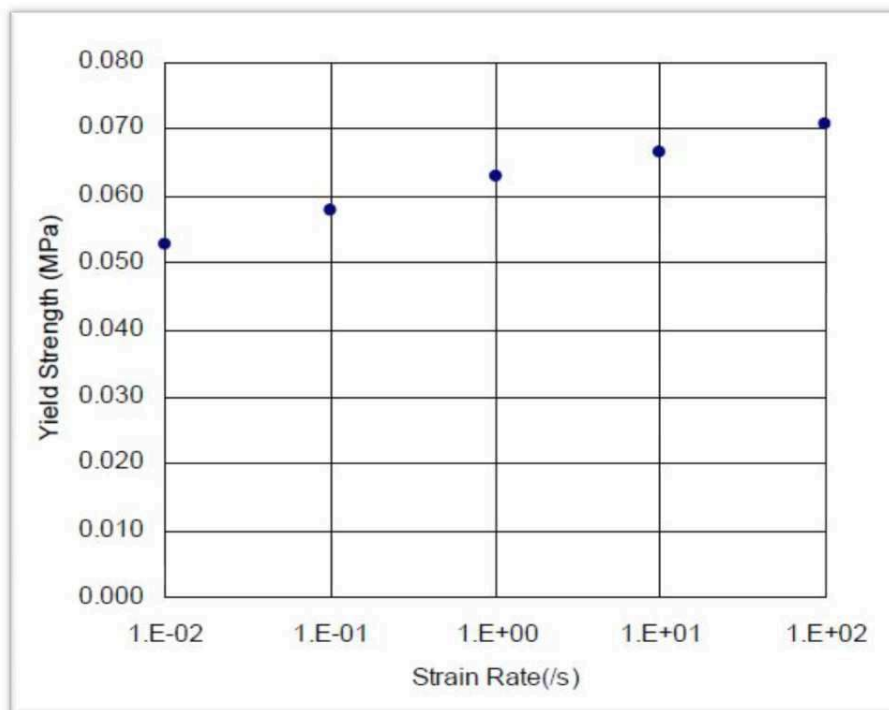


Figure 2: Strain rate dependency of the yield strength of crushable EPS foam.

2.2 General Behavior Crushable Foams

The complexity of the mechanical behavior of the crushable foams is a consequence of its cellular structure. Compression is the most common mode of deformation for crushable foams as they are weak in tension and shear. However, tension and shear deformation can occur due to the geometry of crushable-foamed components. In pure tension, the response of the material is nonlinear only in large deformation.

In pure compression, there are three regions in the stress-strain relationship as shown in Figure 3: Linear compression (cell bending), stress plateau (elastic buckling), and non-linear compression (material densification). Initially, the compressive stress-strain relationship shows a linear increase corresponding to uniform elastic compression where the foam where the crushable foam may have some stiffness due to the strength of the matrix material itself.

The following region is an extended post-yield plateau, in which the collapse of cells spreads through the structure. Pure compression of many foam materials results in the “strain softening” behavior which is described as the drop in the stress at the beginning of the plateau region. In this region, the gaseous component is affected and compressed inside the closed cells of the crushable foams. The gas may be released to the atmosphere if the pressure gets high enough to rupture the cell. This results in an irrecoverable rupture to the cell and permanent damage to the crushable foam material. However, if the matrix material is strong enough, the cell remains unbroken but collapse completely.

The final phase of the compression is a significant steepening increase in the stress as a consequence of the total cell collapse or rapture. The compressed material will begin to behave as a homogenous solid. The elastic modulus, yield stress, and yield strain are defined in the first region whereby the plateau stress and densification strain are defined in the plateau region in which the stress remains almost constant [3,4,8].

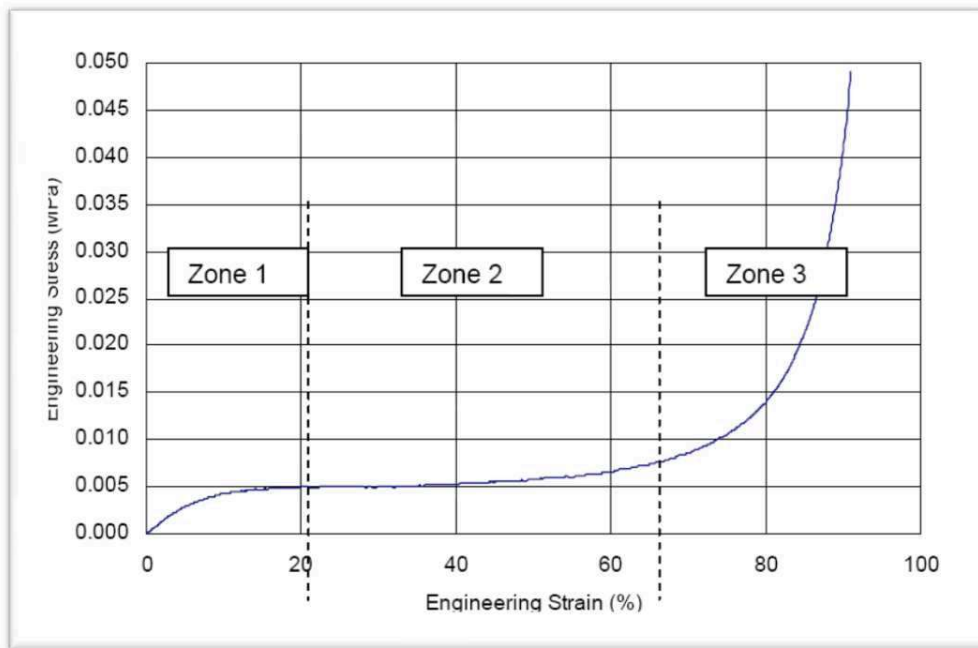


Figure 3: Three deformation zones are typically observed stress strain curve of foams.

2.3 Applications of Crushable Foam

Throughout the past years, the range of applications of crushable foams has become wider and larger as engineers and designers altering the microstructure of the foam materials in order to achieve the desired mechanical properties and behavior that meets the requirements of the applications. They are mainly used in cushioning, impact mitigation, energy absorption, comfort applications. In this subsection, selected applications of the crushable foams are discussed.

One way to increase the car body stiffness and crush-worthiness is the utilization of local reinforcements. The crushable foams have several advantages over other reinforcement materials because of their high absorbing capabilities, low cost and weight. Figure 4 shows the typical reinforcement areas for crushable foams. In this application, BETAFOAM™ is utilized. This foam is synthesized from polyurethane [9]. Figure 5 compares the deformation of un-foamed and foamed car in a frontal crash.

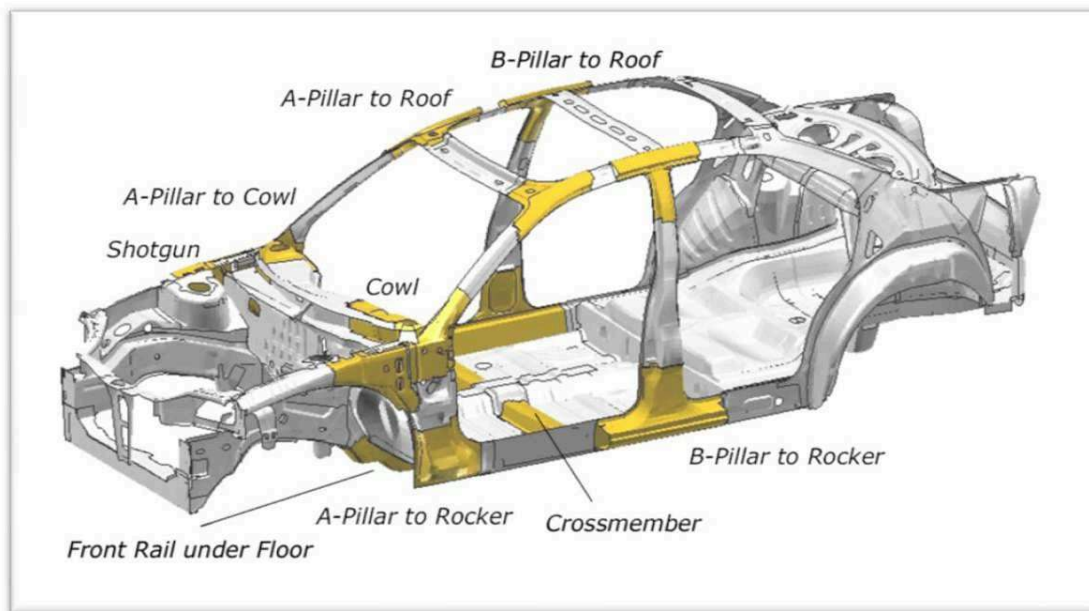


Figure 4: Typical reinforced areas for crushable foams.

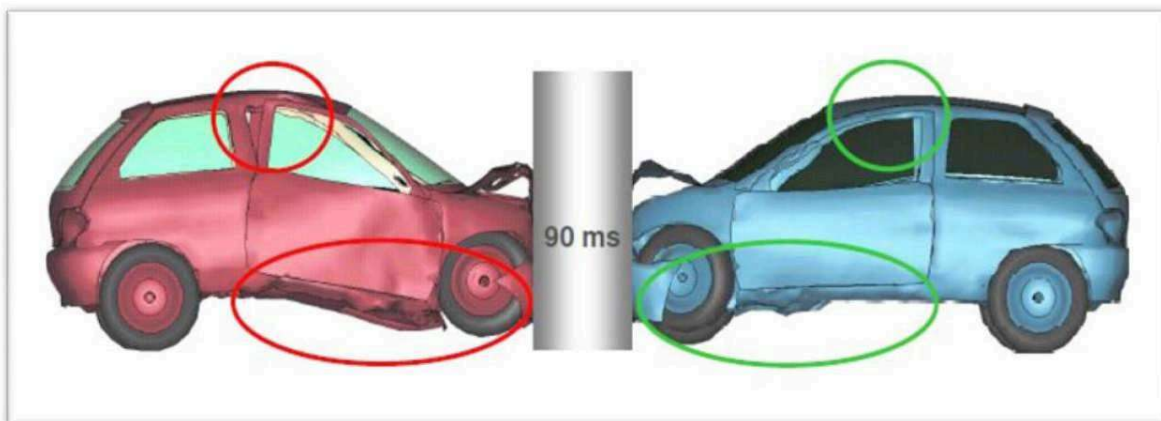


Figure 5: The deformation of un-foamed (left) and foamed (right) car in frontal crash.

Another application for the crushable foams is in oil well casing. Heat is generated due to the normal drilling and production applications. As temperature rises, the trapped fluids tends to expand and potentially can create high pressures which might reach 70MPa. The most effective mitigation solution for this annual pressure build-up is the application of crushable foam wrap. This will allow the fluid trapped within the casing annulus to expand. The crushable foam wrap is predetermined to collapse before any potential dangerous pressure exists. Figure 6 shows how the crushable foam wrap is applied [10].

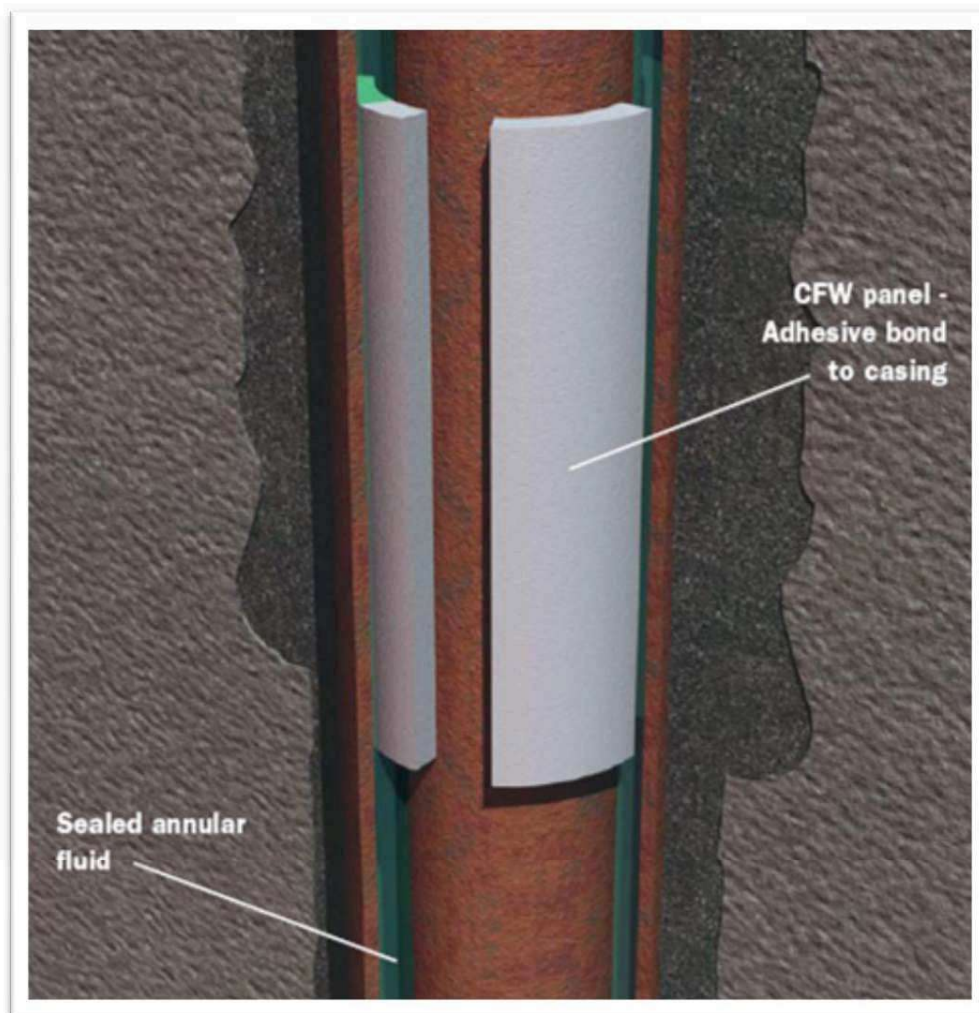


Figure 6: The application of crushable foam wrap.

Another important application of the crushable foam is aircraft runway arrestor systems. The aircraft might overrun the available runway area while takeoff or landing. Overrun occurs in occasions where the aircraft lands “long” on the runway and thus do not have sufficient distance to stop before the end of the runway. Crushable foam arrestor bed systems mitigate the overrun which prevents accidents involving aircraft damage and loss of life [11]. A crushable foam arrestor bed system is shown in Figure 7.



Figure 7: Crushable foam aircraft runway arrestor system.

In automotive safety, the crushable foams are used in the new Steel And Foam Energy Reducing (SAFER) barriers. In many NASCAR racetracks, simple crushable polystyrene insulation foams blocks are placed between the outer steel tube, and the inner concrete wall. This new SAFER barrier is very low in cost and weight and easy to fabricate. Figure 8 and 9 shows an experiment and a simulation of a car crashing with the SAFER barrier, respectively [12,13].

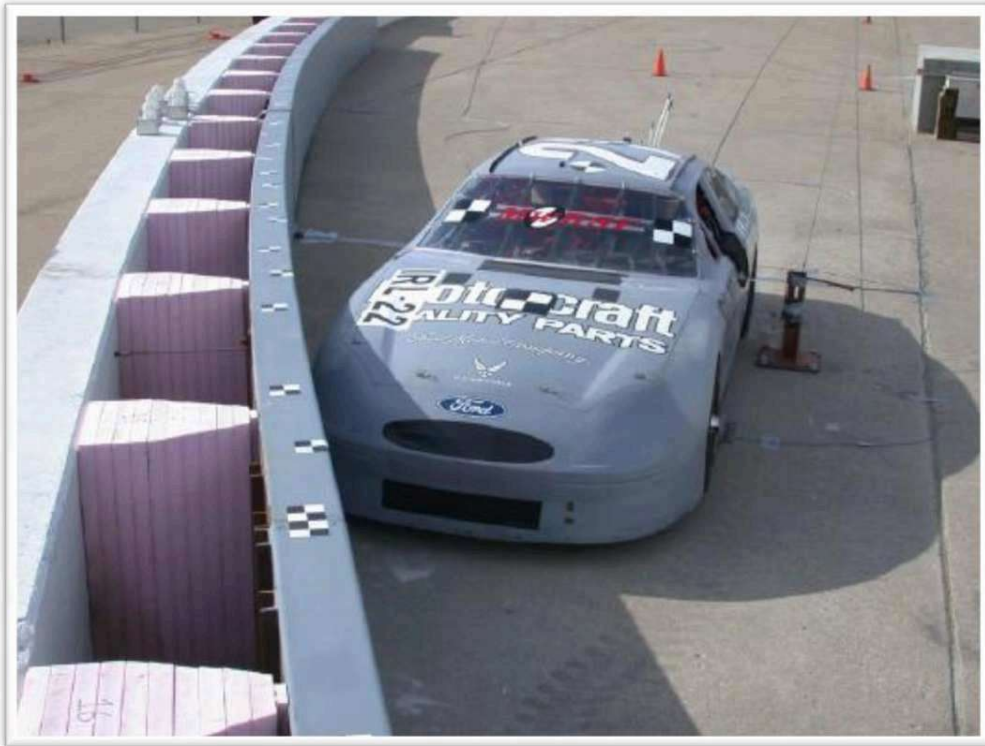


Figure 8: Testing of a car crashing into SAFER barrier.

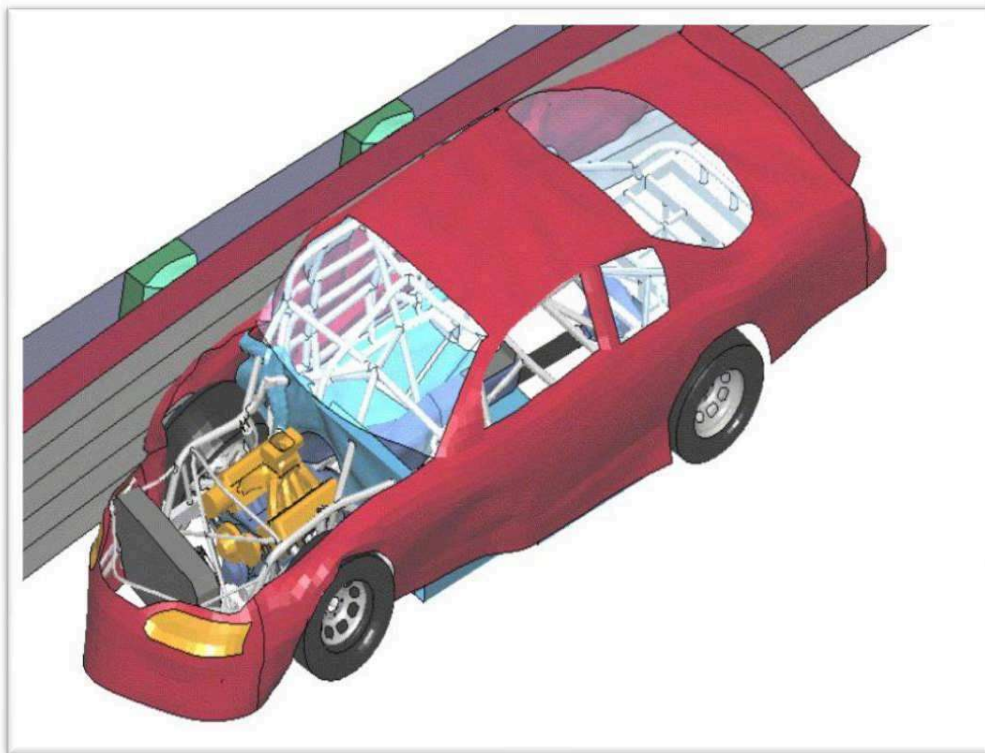


Figure 9: Simulation of a car crashing into SAFER barrier.

CHAPTER 3

EXPERIMENTAL WORK

3.1 Quasi-static Compression Test

The objective of the quasi-static compression test is to find out the stress-strain compression curve of the crushable expanded polystyrene foam. This curve is very important as it is a necessity in making the model for simulation as all foam material models in LS-DYNA require the input of the stress-strain curve. Also, the test was done with different strain rates to study the rate dependency of the stress-strain curve of the expanded polystyrene foam.

The specimens were prepared by cutting a bulk of crushable expanded polystyrene foam, obtained from the engineering automotive lab, into nine identical cubes. Each cube had the dimensions of 10x10x10cm. Four specimens are shown in Figure 10. The density of the foam was calculated experimentally by finding the mass of each specimen then dividing it by the volume. The mass of the specimens was measured by using of the electronic scale as illustrated in Figure 11. The density of the crushable polystyrene foam used for the experiment varies from 11.62 to 13.71 kg/m³ as shown in Table 2. This slight variation of the density is due to the irregularities in the cellular microstructure of the foam. The average value of the density was found to be 12.75 kg/m³.



Figure 10: Cube specimens for the quasi-static compression test.



Figure 11: Measuring the mass of the cubic specimen.

Table 2: The density of the specimens used in compression the experiment

Specimen	Density (kg/m ³)	Specimen	Density (kg/m ³)	Specimen	Density (kg/m ³)
A	11.62	D	12.70	G	13.35
B	12.66	E	12.30	H	12.60
C	13.71	F	12.40	I	13.40

The quasi-static compression test was performed in three different compression rates. The specimens A, B and C were compressed at 500 mm/min while for the specimens D, E and F the compression rate was reduced to 250 mm/min. As for the specimens G, H and I, the compression test was done at 50 mm/min. In all cases, the specimen was compressed up to 80%. The initial and final conditions of the experiment are shown in Figure 12.



Figure 12: Quasi-static compression test of crushable EPS foam.

3.2 Drop Test Experiment

The quasi-static compression test was conducted to study the behavior of the crushable expanded polystyrene foam under slow strain rates. However, due to the strain dependency nature of the crushable foams, the behavior as well as the energy absorption capabilities will differ at higher strain rates. Thus the drop test was carried out to investigate the response of the expanded polystyrene crushable foam to high velocity deformation. Also, the test was conducted to help the process of developing the material model.

The bulk of expanded crushable foam was cut into three identical specimens for this experiment. Each specimen had a 25x25cm square base and a height of 43 cm. As previously calculated, the average of the density is 12.75 kg/m³. The projectile was a cylinder with a diameter of 5 cm and a total length of 65 cm and it was made of standard aluminum with the density of 2700 kg/m³. One end of the projectile was machined to a smooth semi-circle to resemble the head of a missile.

The equipments used for the drop test experiment were a rectangular steel plate and a 6-meters pipe with a diameter of 10 cm. The setup of the experiment is as follows. The steel plate was placed under the expanded polystyrene crushable foam to avoid possible unwanted damage to the ground. The specimen was positioned in the center of the steel plate. The pipe was fixed vertically above the specimen. This pipe serves as a guide to ensure that the dropped object falls vertically also as a measure of the altitude of the free fall. The missile was dropped through the guide and accelerated due to gravity. The arrangement of the experiment is shown in Figure 13.



Figure 13: The setup of the drop test experiment.

The impact velocity in this experiment is equivalent to the free fall velocity of the missile from the altitude of 6 meters. This velocity can be calculated numerically. Mohazzabi and Shea found that the magnitude of the free fall velocity is as follows [14]:

$$|v| = v_t \sqrt{1 - e^{-2g(z_0 - z)/v_t^2}} \quad (2)$$

From the previous equation, the free fall velocity depends on three parameters. The first one is the terminal velocity or v_t which is the maximum velocity an object can have during a free fall velocity. The second one is the standard gravity or g which equals to 9.80665 m/s^2 . The last parameter is $(z_0 - z)$ which stands for the difference between the initial and final height, in other words, the distance of the free fall.

According to Robenson and Crowe, the terminal velocity of a free falling object is given by the following equation [15]:

$$v_t = \sqrt{\frac{2F_D}{C_D A_p \rho}} \quad (3)$$

The terminal velocity depends on four factors: drag force, the density of the fluid medium, the drag coefficient and the projected area of the object. Since there is no acceleration when the terminal velocity is achieved, the drag force is equivalent to net weight of the object. The mass of the object is needed to find its net weight. The mass of the object was calculated from finding its volume, as the density is known. All the dimensions of the missile were measured and its volume was calculated as 1488.984 cm³. Multiplying this value of the volume by the density of the missile, the mass was found to be 4.02 kg. Finally, the net weight of the missile was calculated as 39.425 N.

The experiment was done under normal atmospheric conditions, and the density air was 1.18 kg/m³. Since the diameter of the missile is known, the surface area can be calculated directly and was found to be 19.635 cm². The drag coefficient is 0.99 for long cylinders with the length to diameter ratio equivalent to or larger than 8 [15].

After finding all the parameters, the terminal velocity of the missile was found by substituting them into the previous equation. It was found to be 184.184 m/s. Substituting this value in the free fall velocity equation, the impact velocity of missile was found to be 10.838 m/s.

CHAPTER 4

NUMERICAL SIMULATION

4.1 Material Model Selection

LS-Prepost has a huge library of material models and choosing the right one for the simulation is a difficult task. According to previous works, the best material model for the crushable expanded polystyrene foam is the material model 63 *MAT_CRUSHABLE_FOAM [3,12,13]. The density entered in the input card was the calculated average density. The modulus of elasticity obtained from the compression test was used. As founded in the literature and confirmed by the compression test, the Poisson's ratio is zero. LCID stands for load curve ID of the stress versus strain. The obtained curve from the compression test at highest compression rate was used in the simulation. TSC stands for the tensile cutoff stress which is maximum stress in tension and basically it models foam behavior in tension [16]. The used value for TSC is 0.1 MPa as found in the literature review. The last parameter is DAMP which is the damping coefficient. The recommended value of DAMP is between 0.5 and 0.05. To avoid the negative volume problem, the maximum recommended value of DAMP was used [17].

A simulation of the quasi-static compression test was developed in order to validate the chosen material model. The crushable foam specimen was modeled with the same dimensions of 10x10x10cm. The model was fixed at one end and being compressed at the other end. This was done simply by creating two sets of nodes, one at the bottom and another at the top. The bottom set of nodes was fixed by applying BOUNDARY_SPC_SET

command. A constant velocity was applied on the other set of nodes, by using `BOUNDARY_PRESCRIBED_MOTION` command which required an input curve. The displacement verses time curve was defined by using `DEFINE_CURVE` command.

4.2 Missile Mesh

The mesh of the missile is important. A smoother mesh is desired to achieve more accurate results. However, increasing the number of elements will lead to longer running time of simulation. Thus, a compromise between the two must be observed. The missile model was designed using CATIA and imported to LS-PrePost. The solid was meshed using the solid mesher command. However, to obtain smoother and more uniform mesh, the missile elements were created manually using LS-PrePost. First, a sphere with the missile diameter was meshed, the upper half was removed and then the resulting surface was extruded. The two meshes of the missile are compared in Figure 14

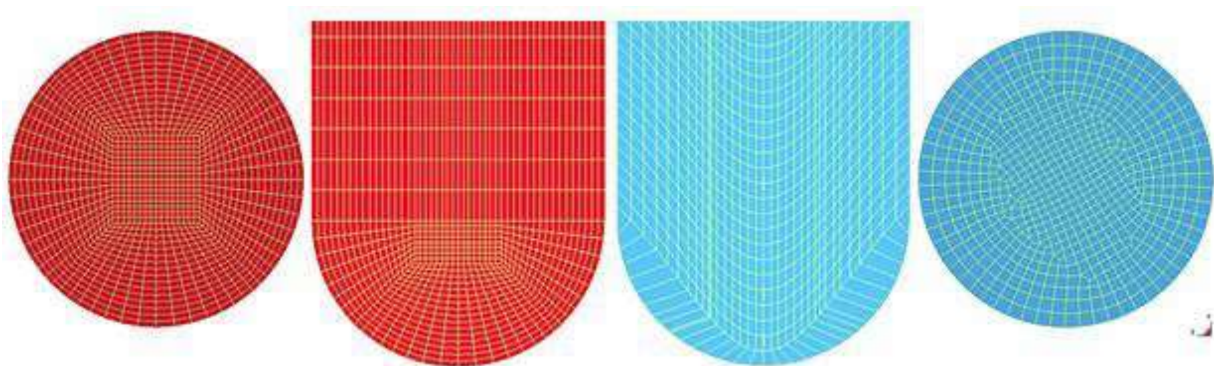


Figure 14: Manual (red) and automatic (blue) mesh of the missile.

4.3 Negative Volume Error

The simulation of quasi-static compression test was straightforward due the uniformity of the compression, the very slow deformation speed. However, while

attempting to simulate the drop test with the same material model, the solution was prematurely aborted due to the negative volume error. This negative volume error is a common problem that occurs in modeling soft materials that undergo large deformations. These materials include soft foams and soils. The distortion of an element might be so severe that the volume of the element is calculated as negative. There are many precautionary measures to eliminate this problem. One of them is to increase DAMP to the maximum recommended value of 0.5, as mentioned earlier.

It may occur that the strain in the simulation exceeds the last point of the stress-strain curve. If this happens, LS-DYNA extends the curve linearly with last slope of the curve. This might lead to relatively small stress values at larger strains and the negative volume problem might occur. Thus, one approach to prevent the negative volume error is to stiffen up the stress-strain curve of the material and extend it exponentially at larger strains. This can be a very effective approach [18]. Figure 15 compares the original and the extended stress-strain curve.

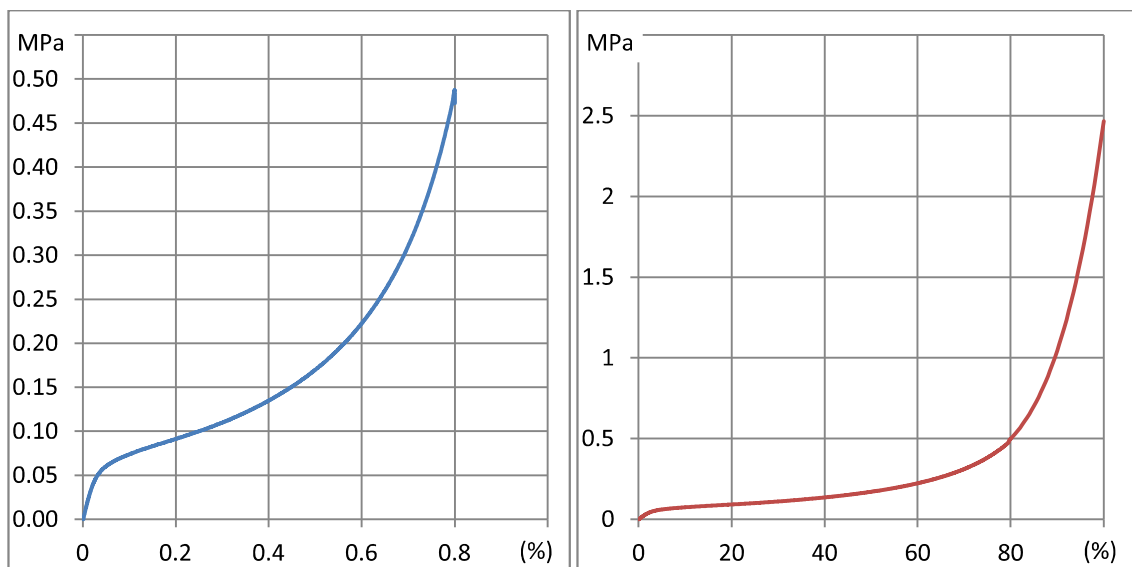


Figure 15: Original (left) and extended (right) stress-strain curve.

Another method of preventing the negative volume problem is to use the constant stress one-point solid element formulation. This is because fully integrated solid elements are less stable during large deformations. However, when using the under-integrated solid formulation, hourglass mode of deformation occurs [19]. Thus, if the constant stress solid formulation is used, the hourglass formulation must be used to obtain more realistic results.

With one point solid integration formulation, the recommended hourglass control type or THQ is 2 or 3, for high velocity impacts, and 4 or 5 for low velocity impacts with the default value of the hourglass coefficient QM [17]. The argument IBQ stands for the bulk viscosity type and it is necessary when severe shocks lead to instabilities. Q1 is the linear bulk viscosity coefficient and it is used to smear the shocks and prevent the element from collapsing at very high velocities. The next argument Q2 is the quadratic bulk viscosity coefficient. This coefficient is important to damp the oscillations [20]. Since there were no instabilities caused by severe shocks in our simulation, these three parameters are left to their default values.

The last two arguments are QB/VDC and QW which are defined as hourglass coefficient for shell bending and hourglass coefficient for shell wrapping, respectively. Since no shell sections were used in the model, these two parameters are ignored.

Another important procedure to prevent the negative volume error is to utilize the contact interior. This type of contact was designed particularly to for soft materials. To define the contact interior, PART_SET of the soft material parts must be created. There are

four important attributes of PART_SET. These attributes can be defined from SET_PART_LIST command.

The first attribute is the penalty scale factor PSF. The second attribute is activation factor. The algorithm starts to act when the compression of the element reaches the activation factor times the initial thickness of the element. The recommended value is 0.1. The third attribute is an optional modulus for interior contact stiffness. The fourth attribute is the formulation type of the interior contact. Type 1 is recommended for uniform compression while Type 2 is designed to control combined modes of compression and shear [17,21,22].

Another approach to avoid the negative volume is to reduce the time step scale factor to avoid numerical instabilities. This is done by changing the value of TSSFAC argument in the CONTROL_TIMSTEP option.

4.4 Material Failure

Crushable foams have a very high energy absorption capability in compression but it is very weak in shear or tension loads. Even during an impact test, the foam might undergo shear or tension due to complexity of the objects shape. Inspecting the result of the drop test simulation in Figure 16, large deformation in the surface was observed. This severe deformation was not observed while conducting the experiment. This unrealistic deformation is because the material model MAT_CRUSHABLE_FOAM does not allow element failure and erosion. The behavior of the crushable foam model in the simulation is similar to that of recoverable foams.

To define the failure criteria, the material card MAT_ADD_EROSION must be used. There are twelve different failure criteria which can be defined. However, for the material model MAT_CRUSHABLE_FOAM only two criteria must be defined. The first failure criterion is due to tensile stress [22]. To define it, MNPRES failure criterion must be used, which is the minimum pressure at failure. The failure will occur if the pressure of the element is less or equivalent to P_{min} . It is very important to note that the pressure is positive in compression and negative in tension. Thus, by setting a negative value for P_{min} , the maximum tensile stress is defined. Any element that reaches this value will be eroded.

The second failure criterion is the plastic strain [11]. This is defined directly by setting the value of MXEPS, which stands for the maximum effective plastic strain. Since two failure criteria were defined, they are calculated independently. If any of them is satisfied, then the failure will occur. In order to combine both failure conditions, NCS was set to 2. NCS indicated the number of failure criteria needed to be met prior to the element deletion. Thus, both of the failure criteria must be satisfied before the erosion occurs.

EXCL is the exclusion number and it is set to a unique value. If any of the failure criteria is equal to the exclusion number, then it is ignored. All the failure conditions are set to the exclusion number except MNPRES and MXEPS.

To find out the values of both failure criteria, a comparison was made between the slow motion video of the experiment and the simulation before applying ADD_EROSION. From this comparison, the point at which the foam starts to fail in the experiment was approximated. Using the Fcomp command, the values of the effective plastic strain and the minimum pressure were found.

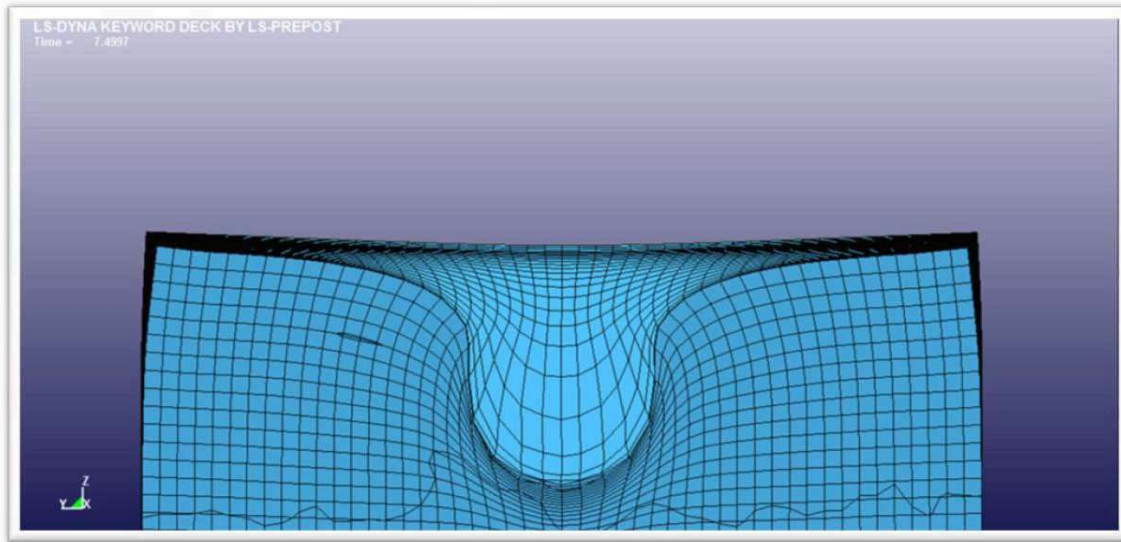


Figure 16: Unrealistic Deformation at the surface of the foam.

4.5 Contact Definition

The contact between the foam and the missile causes instabilities due to the large difference between the stiffness of the two parts. To solve this problem, the optional contact cards A and B must be activated. The value of SOFT is changed to 1 which removes the dependency of the elastic modulus of the parts of the contact by using a self-constraint formulation. The SLDTHK is the optional solid element thickness. This parameter is set any value larger than 1 to prevent penetration between the parts at large strain values and to make sure that no nodes are released from the contact [18].

Because MAT_ADD_EROSION was used, the crushable foam is expected to fail and some of the elements would erode. Thus, some of the inside elements will be exposed to the missile. Surface to surface contact was avoided as it only defines the contact between the nodes of the surface areas of the part. After trial and error, the most suitable contact was

found to be `AUTOMATIC_NODES_TO_SURFACE`, in which a set of nodes containing all the nodes of the foam must be defined.

4.6 Other Commands

Up to this point, the model will run smoothly and no problems will occur. However, in order to obtain more realistic results, some parameters must be taken into consideration. One of these considerations is the mass of the eroded elements. By default, LS-DYNA removes the mass of the eroded elements to make the calculation more stable. However, the erosion is important in our simulation and the reduction of mass might lead to incorrect results. Thus, to prevent mass of the eroded nodes from being eroded, `ENMASS` in `CONTROL_CONTACT` is changed to 1. By doing so, the mass of the eroding nodes of solid elements are retained and continue to be active in contact. The other arguments of this card are kept to their default values [21,23].

Since the energy absorption of the crushable foam is to be investigated, the activation of `CONTROL_COMMAND` is important to get more accurate results. This command provides controls for energy dissipation options. All the four options were set to 2 to include the calculation of dissipations of these four energies: hourglass energy, stonewall energy, sliding face energy, and damping energy dissipations [21].

CHAPTER 5

DISCUSSION

3.3 Results of Quasi-static Compression Test

The quasi-static compression test was conducted to obtain the stress-strain curve and to investigate the strain rate dependency of the crushable expanded polystyrene foam. The stress-strain curve is very important as it is required in modeling the crushable foam materials. The compression test was carried on at three different compression rates. And to obtain reliable results, the test was carried out three times at each compression rate. The data obtained from this experiment are illustrated in Figures 17-19.

The curves obtained from the experiment resemble the ones found in the literature, Figure 1. To make a closer comparison, the data in the literature were converted into numbers using an open source digitizing software. The digitized data is re-plotted along with the experimental data on the same X-Y axis, as depicted in Figure 20. The similarity between the two curves shows the validity of the results of the compression test. The strain rate of the literature curve is 0.1/s while it is 0.083/s for the experiment curve. This little variation in the strain rate is believed to be one of the reasons for the small difference between the two curves. Another reason is the difference in the density of the crushable foams. The stress strain curves in the literature were resulted from compressing crushable foams with the density of 19.6 kg/m³. In the quasi-static experiment, the average density of the crushable foam was 12.75 kg/m³.

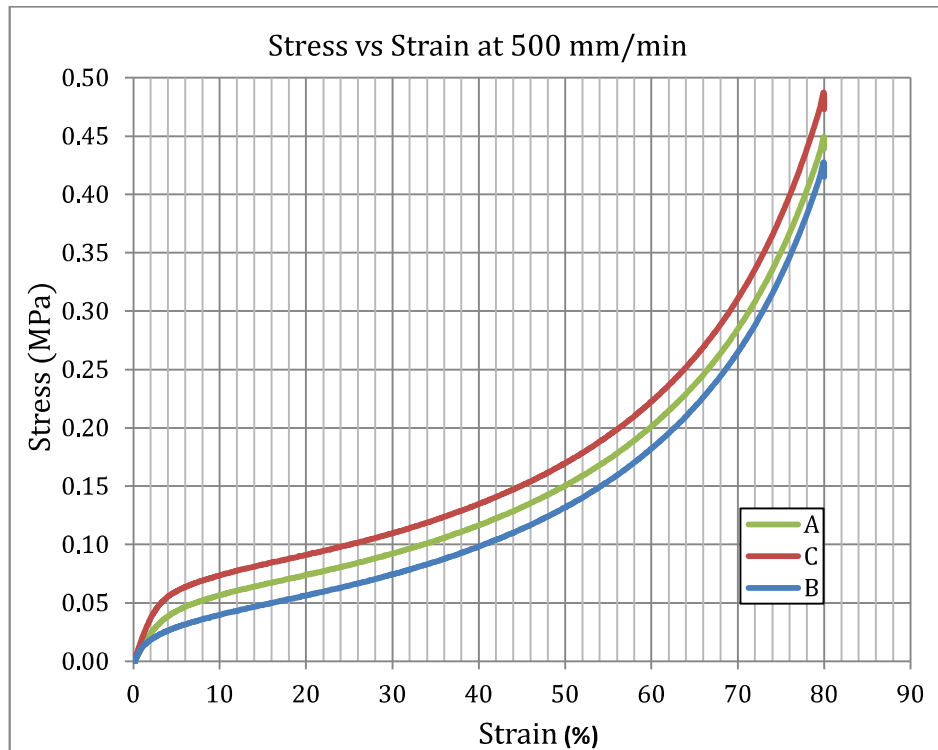


Figure 17: The compression curve of crushable EPS at 500 mm/min.

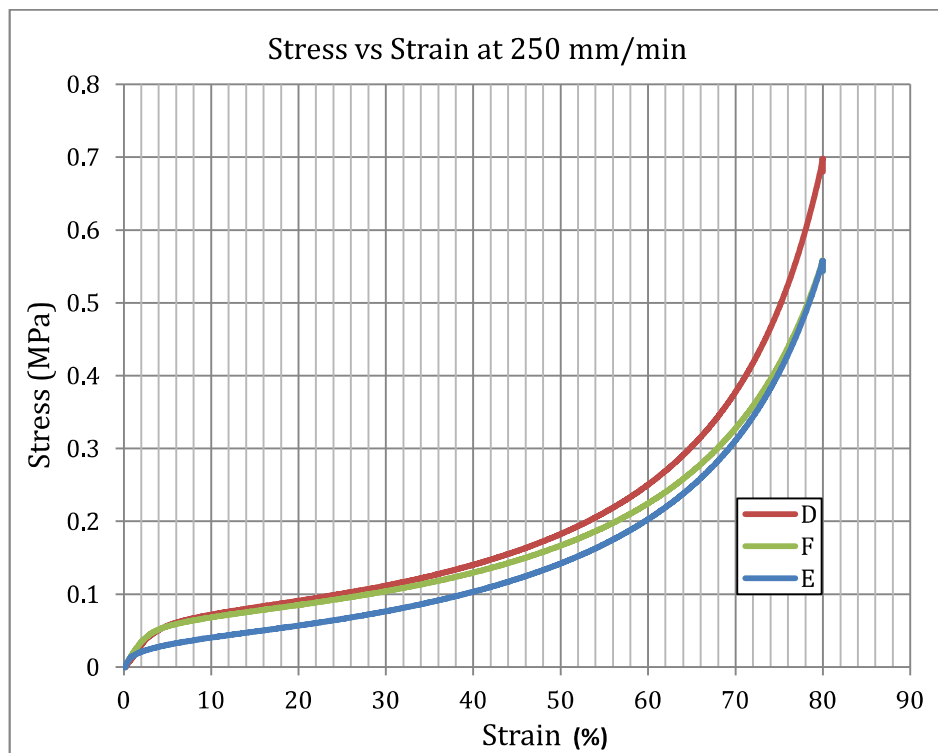


Figure 18: The compression curve of EPS at 250 mm/min.

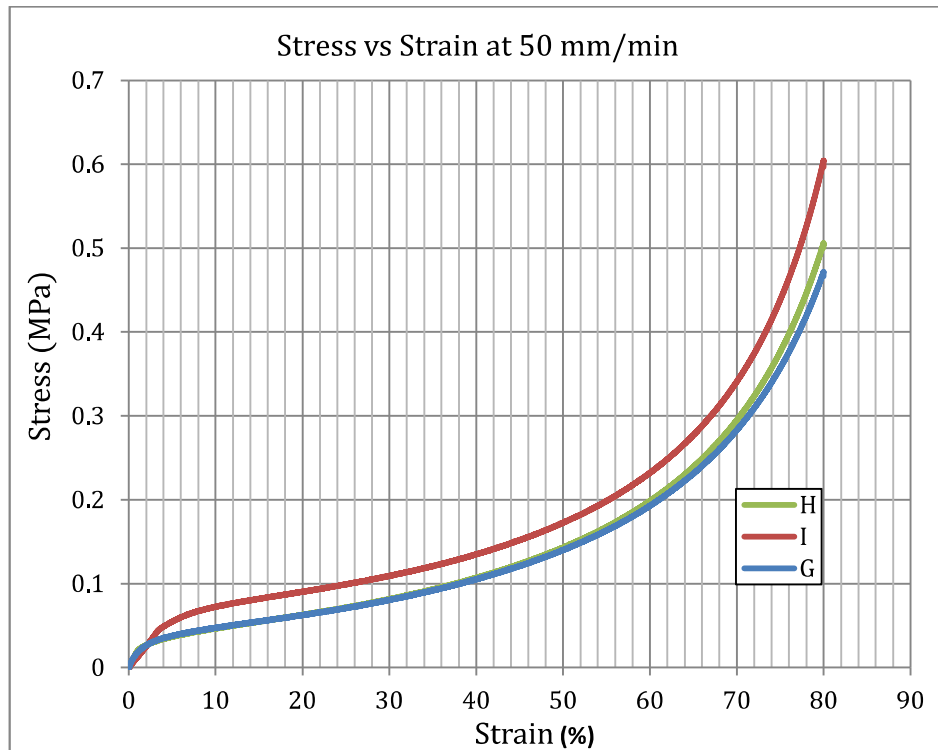


Figure 19: The compression curve of EPS at 50 mm/min.

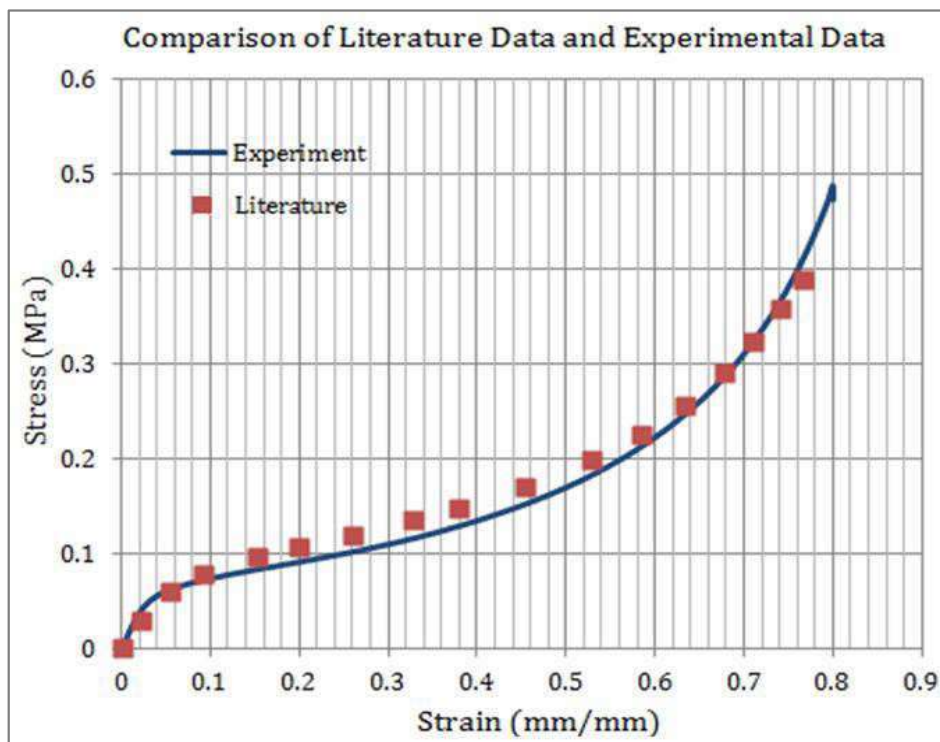


Figure 20: Comparison between literature data and experimental data.

The results of the compression test for each strain rate were averaged. The average of each curve was plotted along the same axis to investigate the effect of the compression rate on the stress-strain curve. The strain rate can be found directly by knowing the compression rate or the speed of deformation and the initial length. In this experiment, the strain rates for the compression rates 50, 250, and 500 mm/min were 0.00833, 0.04167, and 0.0833/s, respectively. As observed from Figure 21, as the compression rate or strain rate increases, the behavior of the crushable foam becomes stiffer. It can be concluded that at higher velocities the crushable foam is able to absorb more energy. This feature makes the crushable expanded polystyrene foam a very good candidate as an energy absorption material in high velocity safety applications. The total energy absorbed was found by calculating the area under the stress strain curve. At 500 mm/min the foam absorbed 135.061 Joules while at 50 mm/min it absorbed 104.883 Joules.

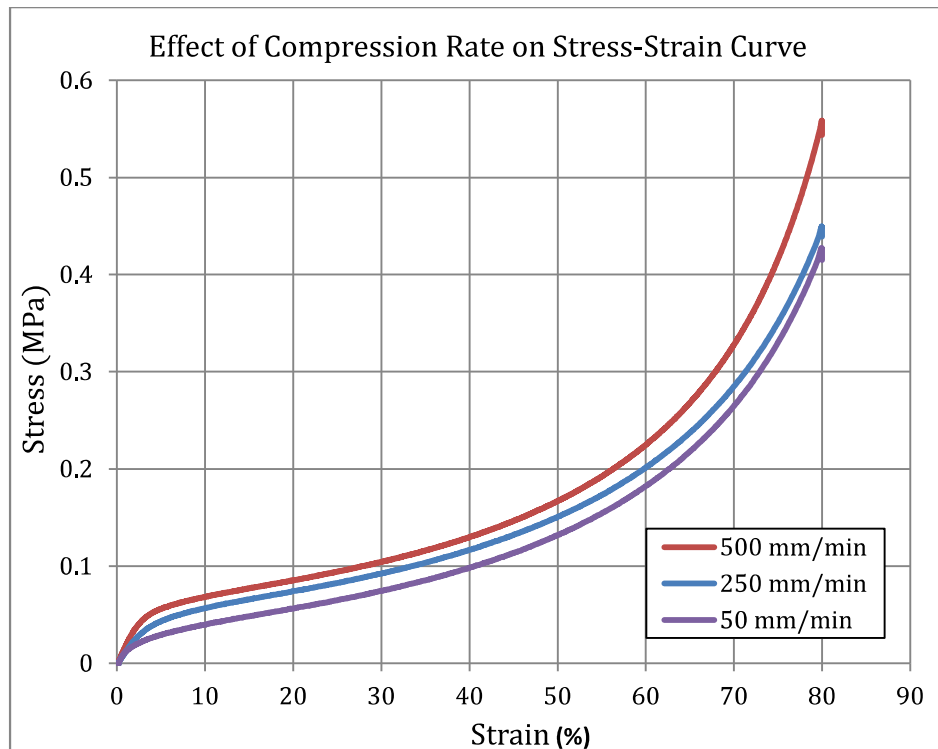


Figure 21: Rate dependency of crushable EPS foam.

The quasi-static compression test was captured by a recording device and the screenshots of the video are displayed in Figure 22. Upon close examination of the video, it was noted that there was no change in cross section of the crushable foam. Thus, as found in the literature, the Poisson's ratio is zero. This implies that the volume of the crushable expanded polystyrene foam is not conserved during compression and its density increases as it is being compressed, similar to gaseous materials. This is due to the existence of the air in cells of the crushable foam.

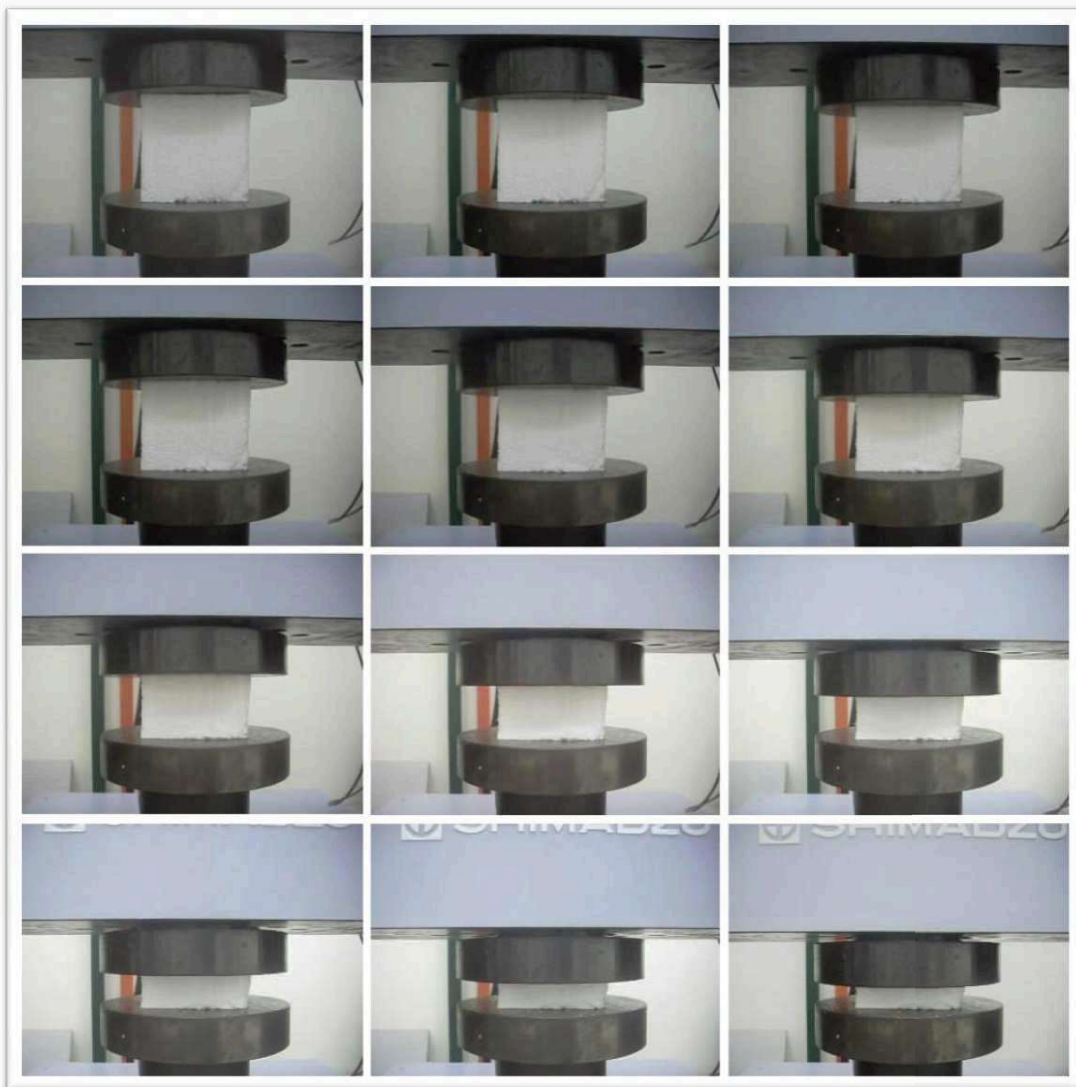


Figure 22: Screenshots of the quasi-static compression test.

Because of zero Poisson's ratio, the crushable expanded polystyrene foam have identical true and engineering stress-strain curve. That is, because the difference between the two curves that true stress-strain curves takes into account the change in the cross section area. However, in our case, there is no change in the lateral dimensions.

The tangent lines of each average stress-strain curve in the linear elastic region were found, as shown in Figures 23-25. The value of the elastic modulus is equivalent to the tangent. The elastic modulus at compression rate of 500 mm/min was 2.06 MPa. As the rate was reduced to 250 mm/min, the modulus decreased to 1.59 MPa. At the rate of 50 mm/min, the modulus was found to be 1.35 MPa. The value of the elastic modulus of the expanded polystyrene crushable foam in the literature was 4.32 MPa. However, this does not necessary indicate the invalidity of our results. As stated before, the value of the elastic modulus of the crushable foam depends on the strain rate and density.

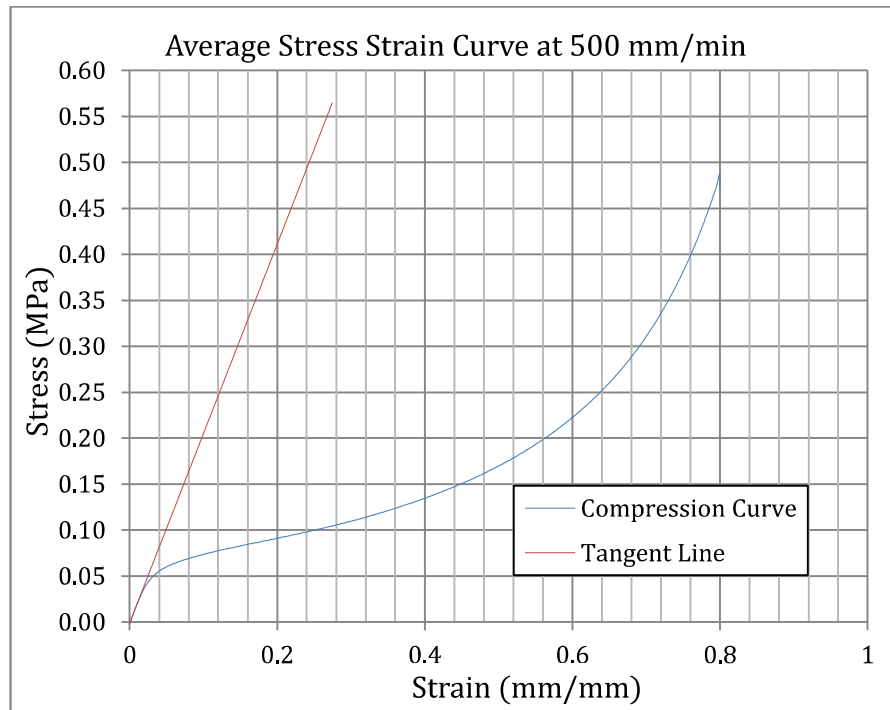


Figure 23: Modulus of Elasticity at 500 mm/min.

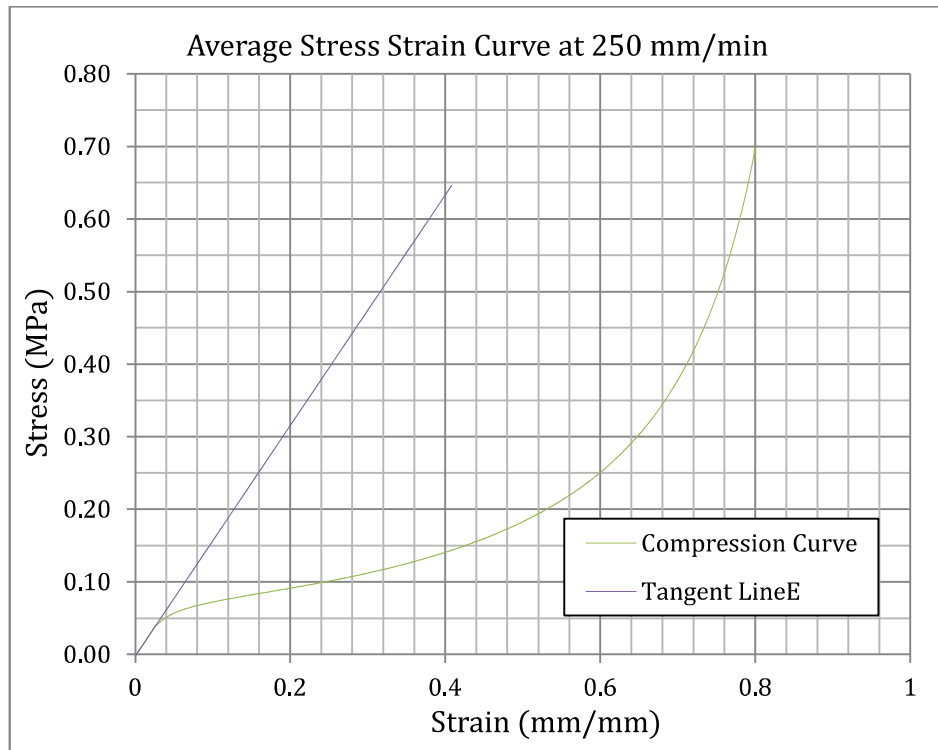


Figure 24: Modulus of Elasticity at 250 mm/min.

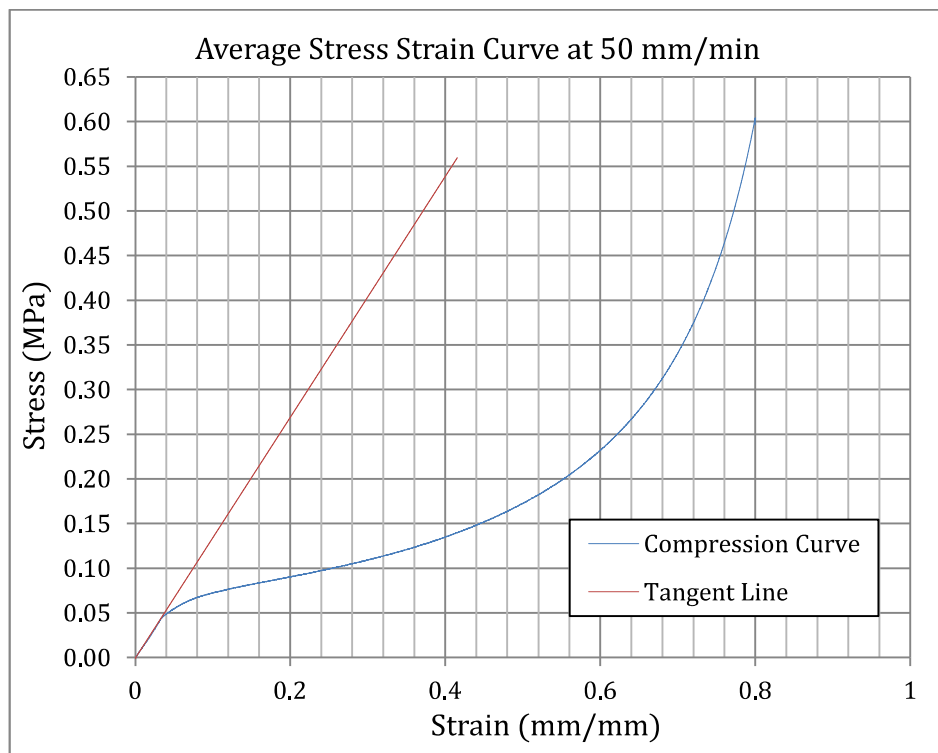


Figure 25: Modulus of Elasticity at 50 mm/min.

5.2 Results of Drop Test Experiment

In the drop test experiment, the compression of the crushable foam was localized in a small area equivalent to the projected area of the missile. Due to the high compressibility, the foam was crushed and there was only little recovery. After the impact, the missile was trapped in the foam specimen as shown in Figure 26. A considerable amount of force was needed to remove the missile from the specimen. This indicates a high friction force. After the localized failure of the material, the surrounded area springs back and exerts inward pushing force on the surface area of the missile. The damage on the surface area of the foam specimen was marked as depicted in Figure 27.

The experiment was captured by a recording device with the recording speed of 120 frames per second. Subsequent frames of the video during the impact are displayed in Figure 28. Inspecting the recorded video in slow motion revealed a slight bouncing back of the missile after it reached the maximum depth of penetration. This was due a small recovery of the foam material. To measure the depth of penetration, the missile was marked before removing it from the foam specimen. The depth of penetration was measure from the tip of the missile head to the mark, as shown in Figure 29. The test was carried out four times to obtain reliable results. The four specimens are shown in Figure 30 and the depth of penetration for each specimen is presented in Table 3.

Table 3: The depth of penetration for each specimen

	Specimen 1	Specimen 2	Specimen 3	Specimen 4	Average
DOP (mm)	189	190	186	187	188



Figure 26: Final position of the missile after the impact.



Figure 27: The damage on the surface of the specimen.

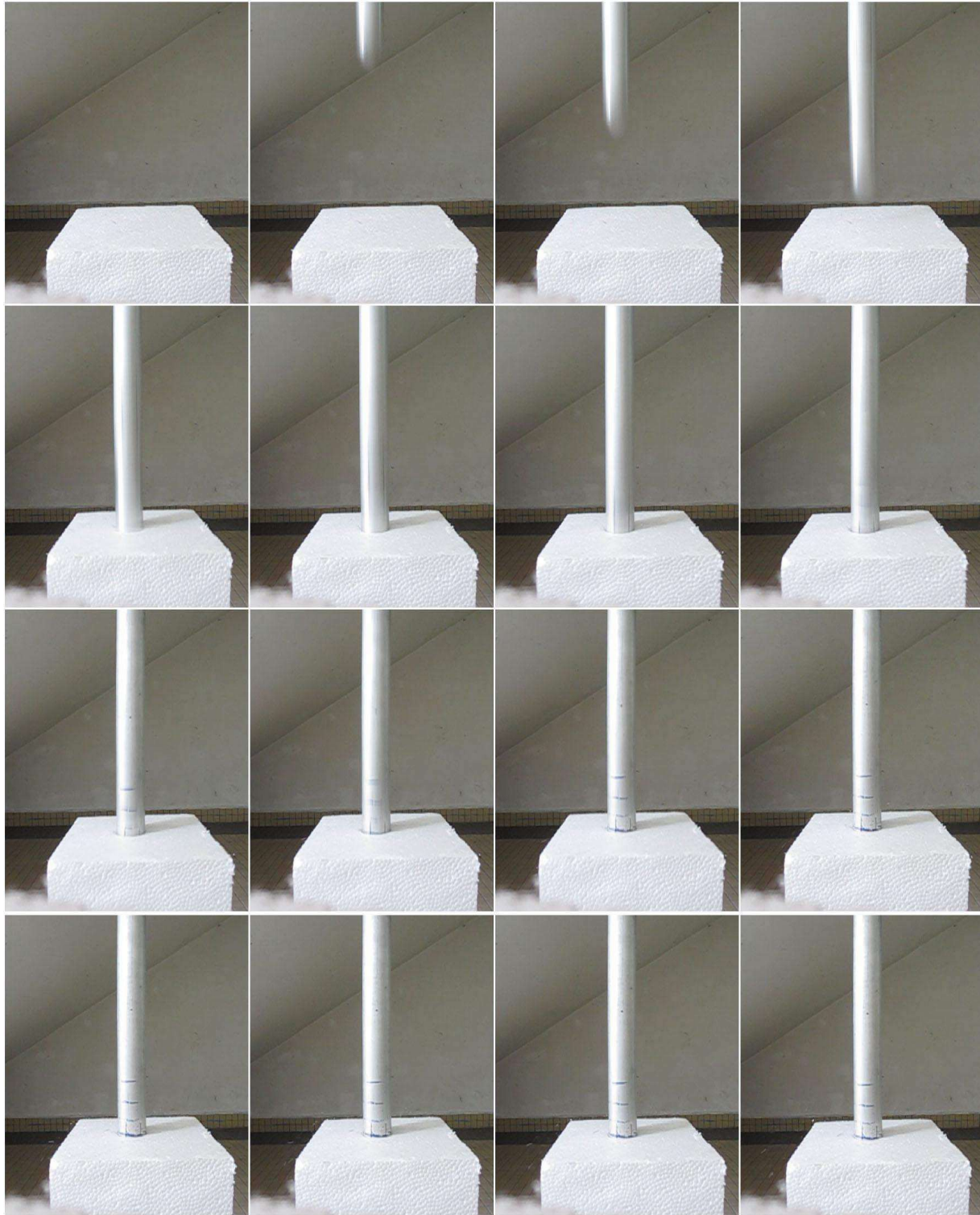


Figure 28: The video frames of the impact test.



Figure 29: Taking the measurement of the depth of penetration.



Figure 30: The four specimens after the impact test

5.3 Results of Compression Test Simulation

The analysis was solved successfully without any errors. The resulting force versus displacement curve was plotted along with the curve which was obtained experimentally. As shown in Figure 31.

The similarity between the two curves in the figures proves the validity of the foam model. Moreover, the screenshots of the simulation which are displayed in Figure 32 are identical to the footage of the captured video of the experiment, Figure 22. Both of the figures show only vertical deformation in the material.

The energy absorbed by the expanded polystyrene crushable foam in the compression test was plotted in Figure 33. The total energy absorbed was 135.85 Joules. Experimentally, the total energy absorbed was 135.061 Joules, which again validates the results of the compression test simulation. With this very small variation in the total absorbed energy, the simulation has a very small percentage error equivalent to 0.7 %.

With a constant compression rate, the absorbed energy increases nonlinearly with the time. The expanded polystyrene crushable foams have zero Poisson's ratio. Thus, the density of the material will increase with compression as there is no lateral deformation. Therefore, more force is required for compression as the material being compressed. As the stress increases exponentially as larger strains, the required compression force is expected to rise exponentially as well.

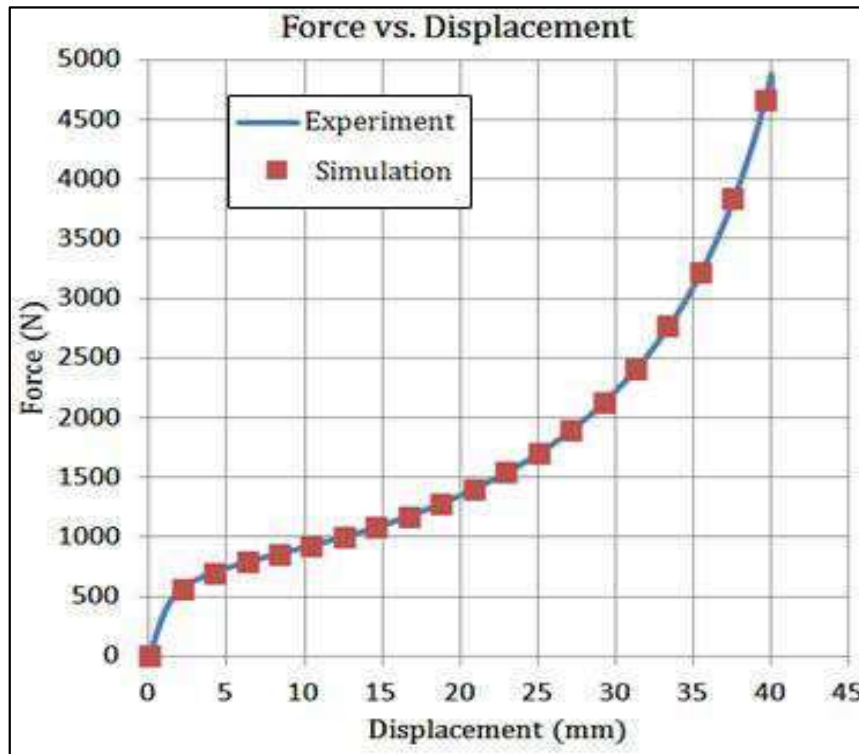


Figure 31: Comparison between experiment and simulation results.

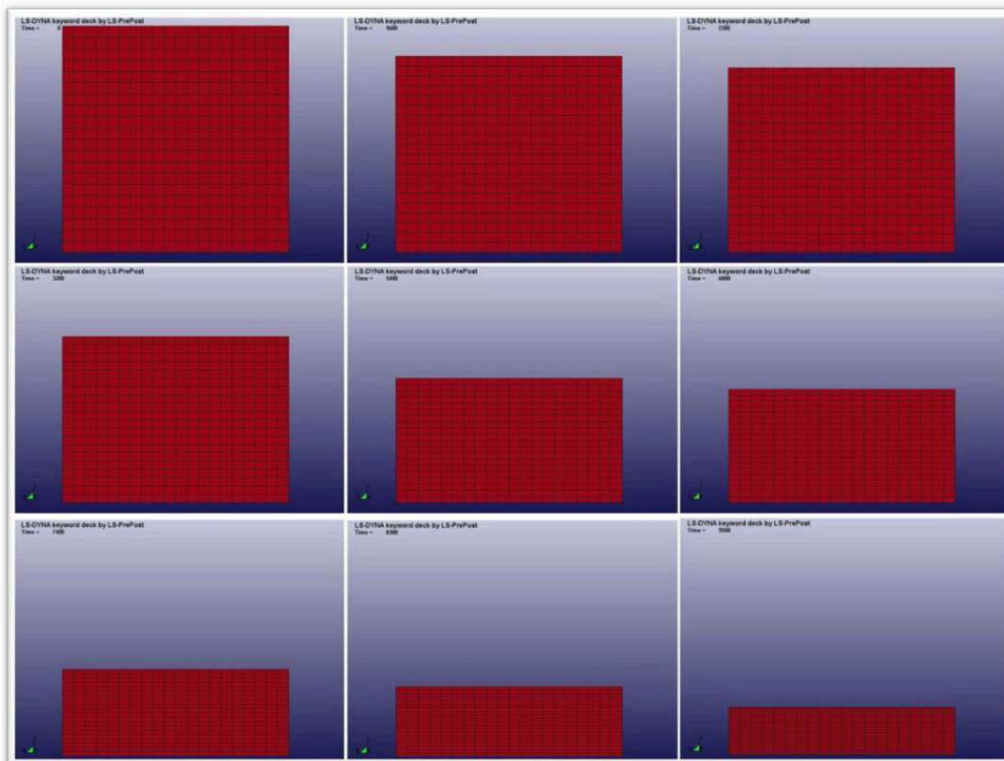


Figure 32: Screenshots of the simulation of compression test.

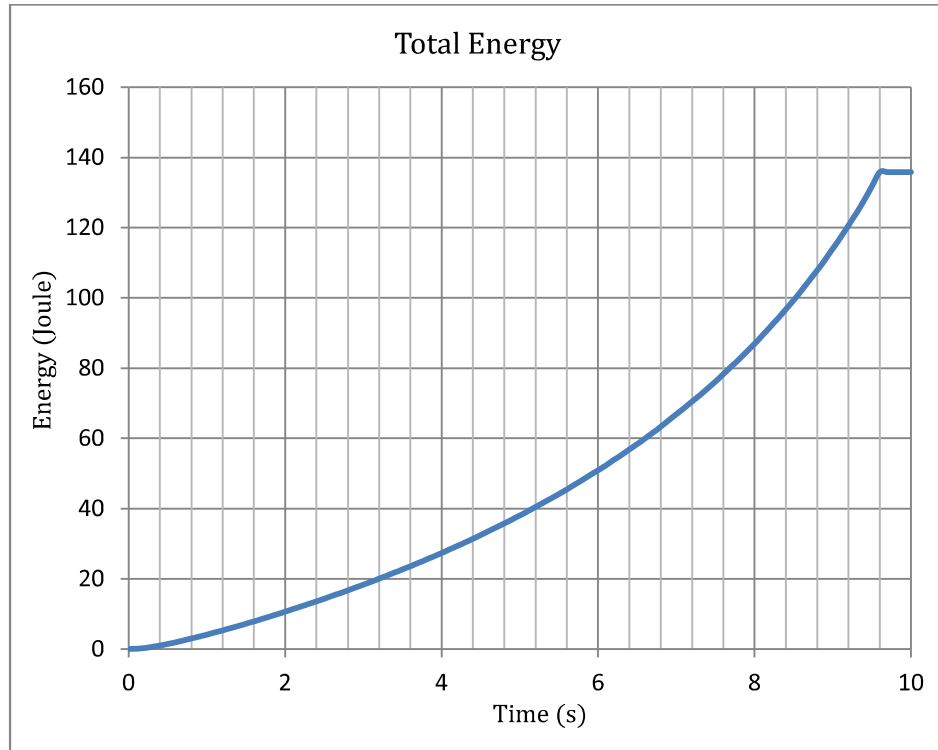


Figure 33: The total energy absorbed by the crushable foam in the compression test.

5.4 Results of Drop Test Simulation

Although the developed model for simulating the quasi-static compression test had negligible errors, it might not be necessary accurate in other cases. One of the reasons was the very slow compression rate of the test which improved the calculations stability. However, the main reason is that the force was applied uniformly to the entire surface area of the specimen in the compression test. In most of the application of crushable foam, the force is applied locally. There will be a combined mode of compression, shear and tension and the crushable foam might be damaged and crushed. This was observed by the large unrealistic deformation in the foam shown in Figure 16. The material model developed in the compression test simulation must be improved.

Investigating the profile of missile displacement in Figure 34 shows that the missile reaches the maximum the depth of penetration of 188 mm at 33 milliseconds. This value is identical to the average depth of penetration in the experiment. It is also observed from Figure 34 that the curve goes up slightly after reaching its maximum value. This is verified by the experimental results as it was observed in the slow motion video of the drop test experiment that the missile bounces back slightly after hitting the crushable foam.

The velocity profile of the center node on the foam surface is plotted in Figure 35. Vibration is noticed in the profile due to the impact. However, the velocity profile of the missile shows no vibrations. This is due to the very high density of the missile material comparatively to the density of the crushable foam.

The energy absorbed by the crushable foam is plotted in Figure 36. The total energy absorbed is 95 Joules. However, in the quasi-static compression test, the total energy absorbed is 135 Joules as observed from Figure 30. This means that the crushable foam absorbed approximately 1.5 more energy. Although the strain rate was lower, more energy was absorbed due to higher compression area. The compressed surface area in the drop test experiment was 19.65 cm³ while the surface area of the specimen in the quasi-static compression test was 100 cm³, approximately 5 times larger.

A comparison between the total energies of the missile and crushable foam is shown in Figure 36. The maximum kinetic energy of the missile is 185 Joules while the maximum energy absorbed by the foam is only 95 Joule. The difference between these two values is caused by many reasons. One of them is the friction force between the missile and the inner surface area of the foam. Another one is the absorbed energy to the ground or rigid wall. It

should be noted that LS-DYNA calculates the total energy as summation of the kinetic and internal energies only.

The failure of the material is shown in Figure 37. This indicates that failure in the foam is similar to that in brittle materials. There will be a crack on the surface area of the crushable foam and as the foam is being compressed, it will propagate horizontally. This was not possible to confirm experimentally as the material exhibited some revocability after the missile was removed.

The damage of the foam was localized in a small area which is the projected area of the missile. This is shown by the plot of the stress in Figure 38. The maximum Von-Misses stress was 35 MPa and it occurs at the center element on the foam surface. Examining the shear strain distribution in Figure 40 clearly reveals that the crushable foam material fails due shear strain as the maximum shear strain occurs at the inner surface area.

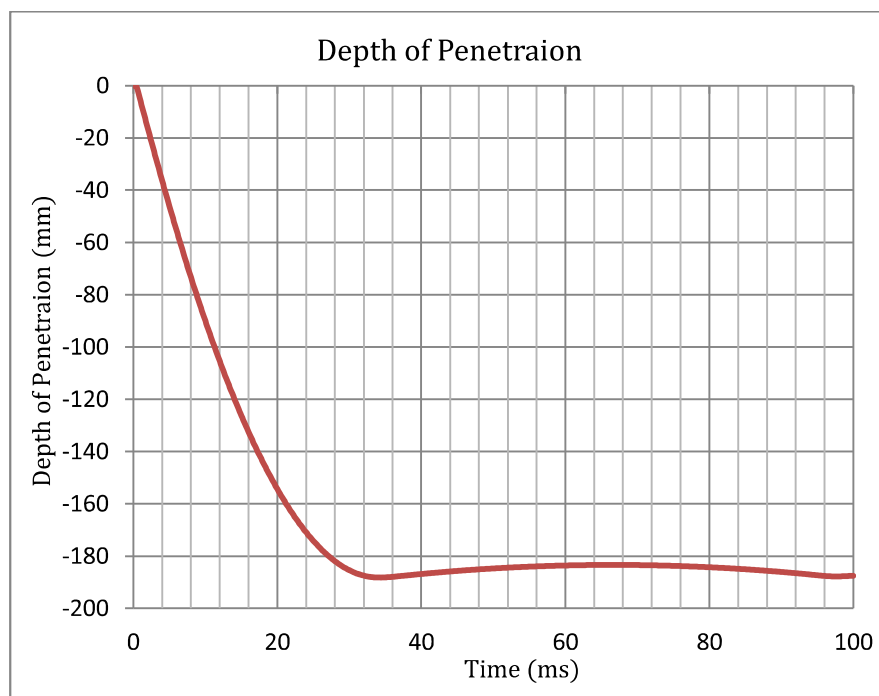


Figure 34: The depth of penetration of the missile in the simulation.

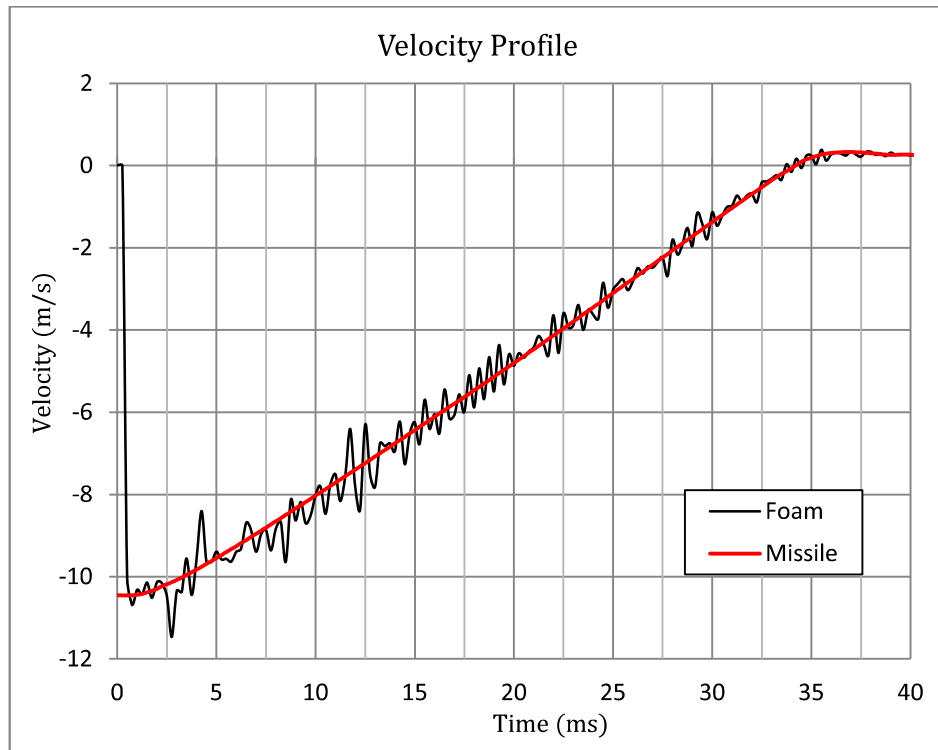


Figure 35: The vibration on the foam due to its low density.

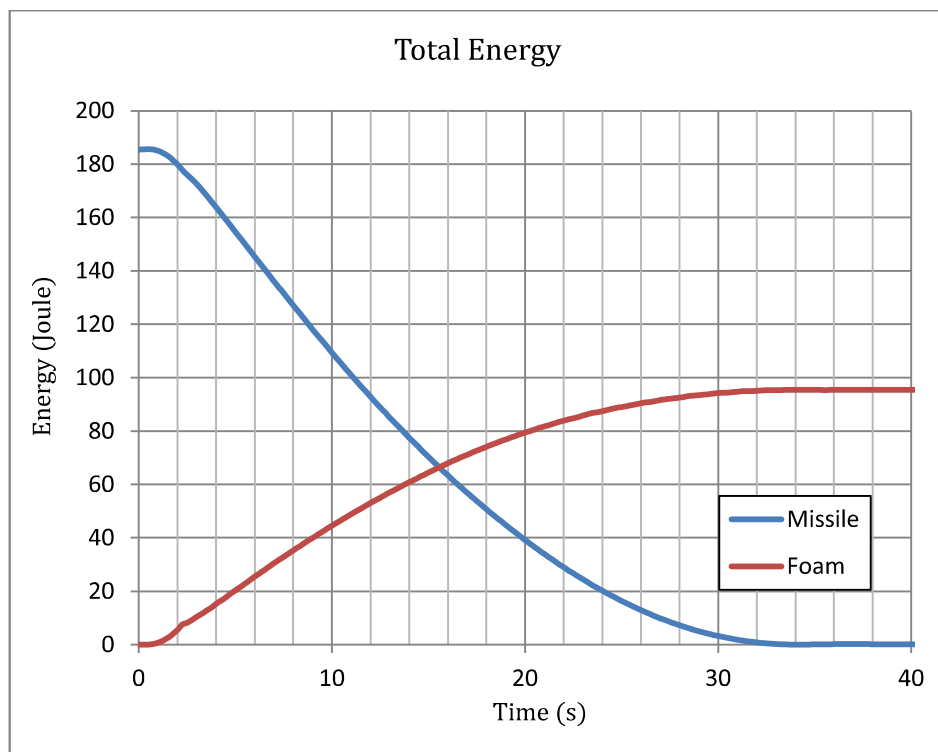


Figure 36: The total energy absorbed by the crushable foam.

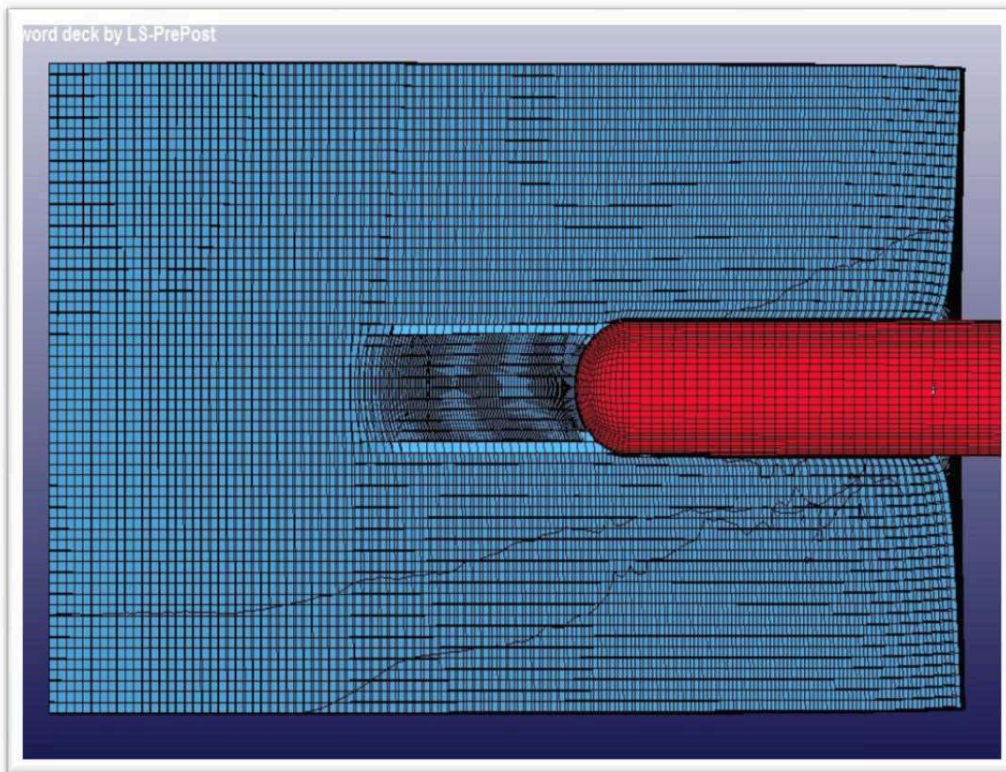


Figure 37: The failure mechanism of the foam elements.

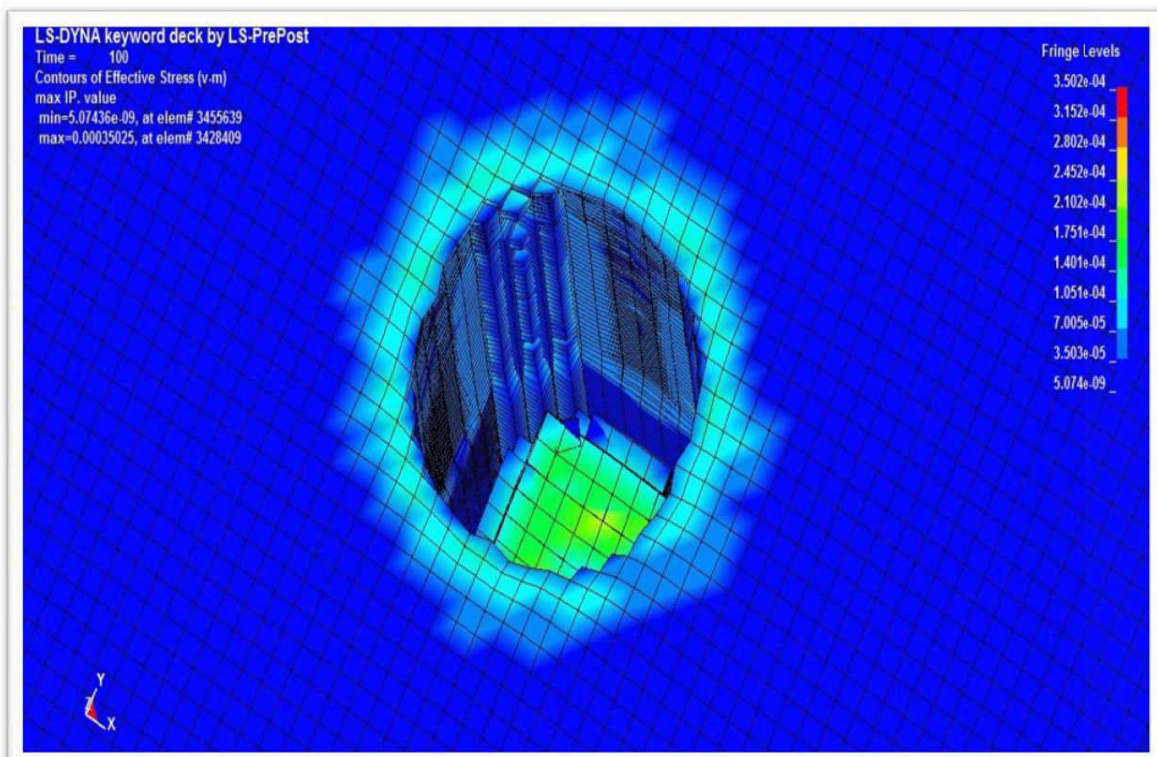


Figure 38: Von-Mises stress distribution in the crushable foam.

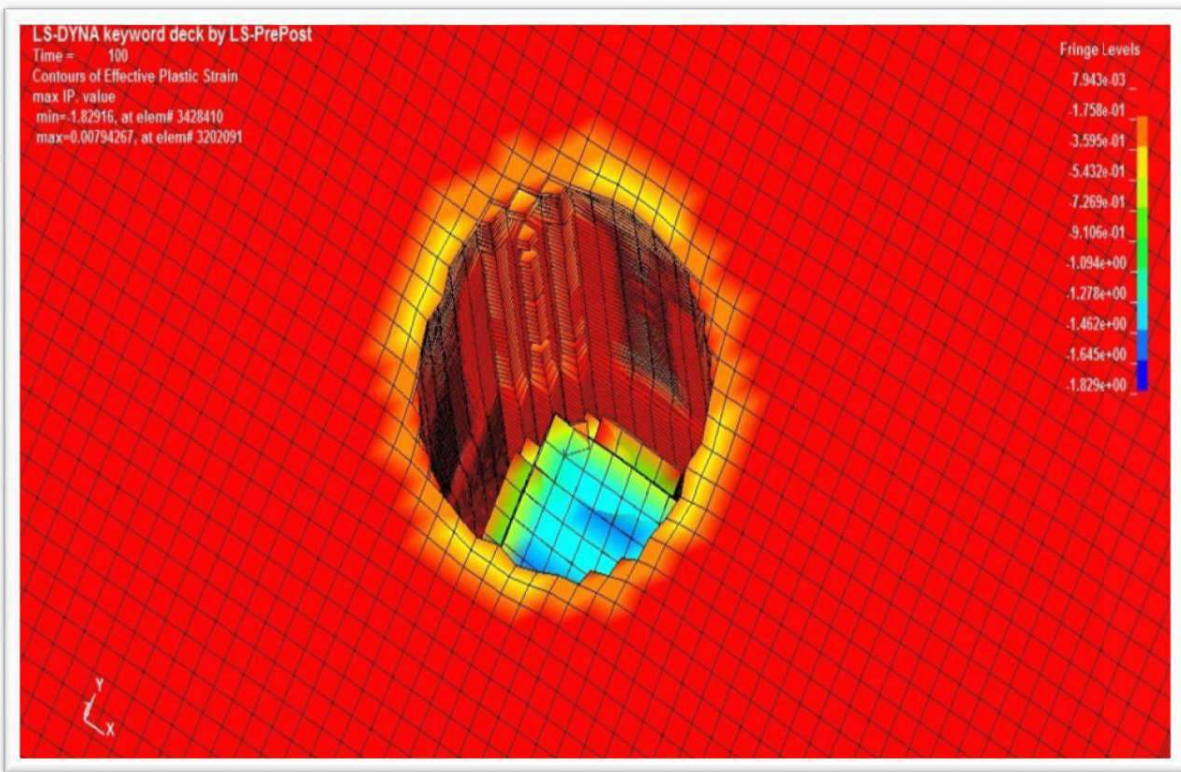


Figure 39: Plastic Strain distribution in the crushable foam.

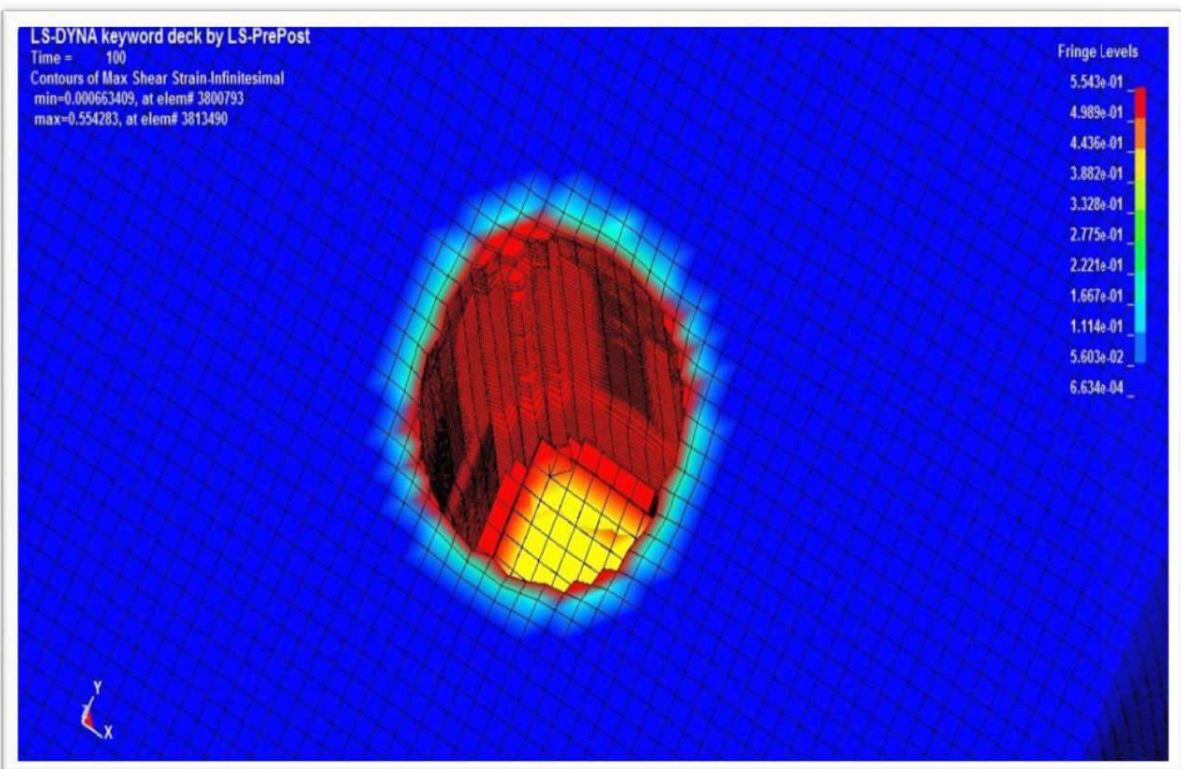


Figure 40: Maximum shear strain distribution in the crushable foam.

5.5 Extrapolating the Foam Model

The developed material model proved to be successful as the results of the simulation match the results of the drop test experiment. Thus, the material model was extrapolated to simulate a car crash in which a vehicle runs into a bulk of the expanded polystyrene crushable foam. The crushable foam had the dimensions of 1.5x1x1 meters and was constrained to a rigid wall. The simulation was performed at five different impact velocities to study the effect of the strain rates on the energy absorption capabilities of the crushable foam. The screenshots for the impact simulation at 10 m/s are displayed in Figure 41.

With the impact velocity of 5 m/s, the vehicle was decelerated to zero approximately after 0.12 seconds from the impact and did not hit the rigid wall after crushing the foam. The average deceleration is 41.67 m/s^2 . It bounces back slightly after the impact as observed in the velocity profile in Figure 42. If the impact velocity is increased to 10 m/s, the total time of impact on the foam is increased to 0.223 seconds and the deceleration is increased slightly to 44.83 m/s^2 .

However, with the impact velocity of 15 m/s, the crushable foam is totally crushed and the vehicle hits the rigid wall. The vehicle comes to zero velocity after 0.128 seconds and the average deceleration is 117.18 m/s^2 . Thus, at this impact velocity, more volume of the crushable foam is required to completely absorb the impact.

To investigate the effect of placing the foam next to the rigid wall in the impact, the profile velocity of the vehicle crashing directly into the rigid wall was plotted in Figure 43. With the impact velocity of 5 m/s, the vehicle reached zero velocity after 59 milliseconds

and the duration of impact is approximately halved. Therefore, by placing the crushable foam next to the rigid wall, the average deceleration is halved.

According to Newton's Second Law of motion, the force is proportionally related to the acceleration. Thus, since the deceleration is reduced by placing the foam, the damage on the car is reduced as well. At the impact velocity of 10 m/s, there is only little deformation and minimal damage on the frontal area for the car due the impact with foam as shown Figure 44. However, if the car crashes the rigid wall directly at the same speed, there is noticeable damage. The deformation of the car is shown in Figure 45.

The effect of impact velocity on impact period and average deceleration is shown in Figure 46. The effectiveness of crashing into the crushable foam decreases as the impact velocity increases beyond 10.75 m/s. If it exceeds this limit, then the car will run through the bulk of the crushable foam and hits the rigid wall. This can be depicted from Figure 46 as the gap between the two impact period curves decreases as the impact velocity goes up.

The crushable foam is a good material in absorbing energy and well used in safety applications. At the low velocity of 5 m/s, it absorbed a considerable amount of energy, nearly 4700 J after being slightly crushed. As we increase the impact velocity to 15 m/s, the total energy absorbed is drastically increases to 14600 J. The total energy absorbed by the crushable foam at different impact velocities is plotted in Figure 47.

These results are validated by quasi-static compression test. As the impact velocity increases, the strain rate increases. Thus, the stress strain curve of the crushable foam becomes stiffer as the impact velocity increases. Therefore, the area under the stress strain curve, which is basically the absorbed energy, becomes larger.

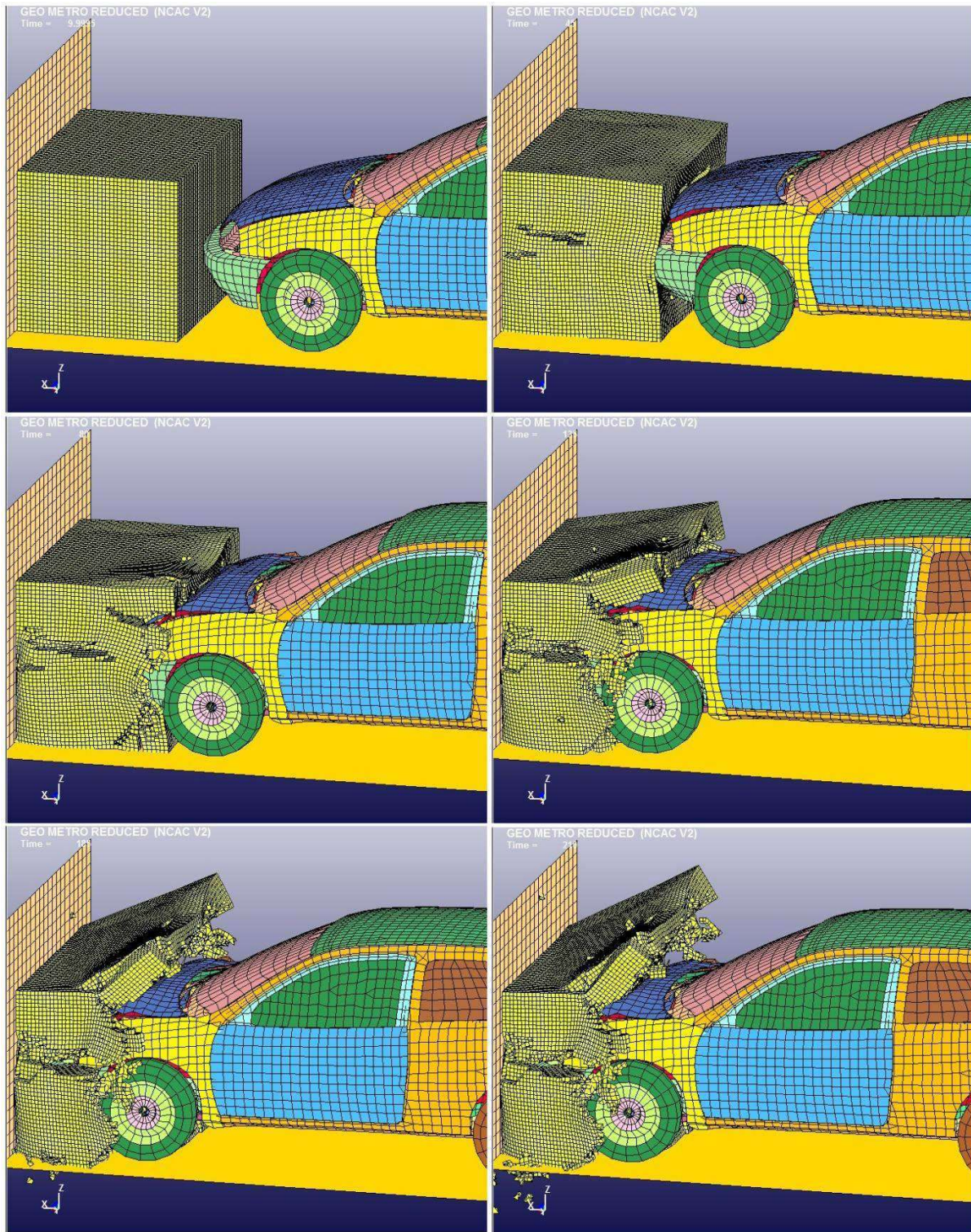


Figure 41: Car crushing foam simulation.

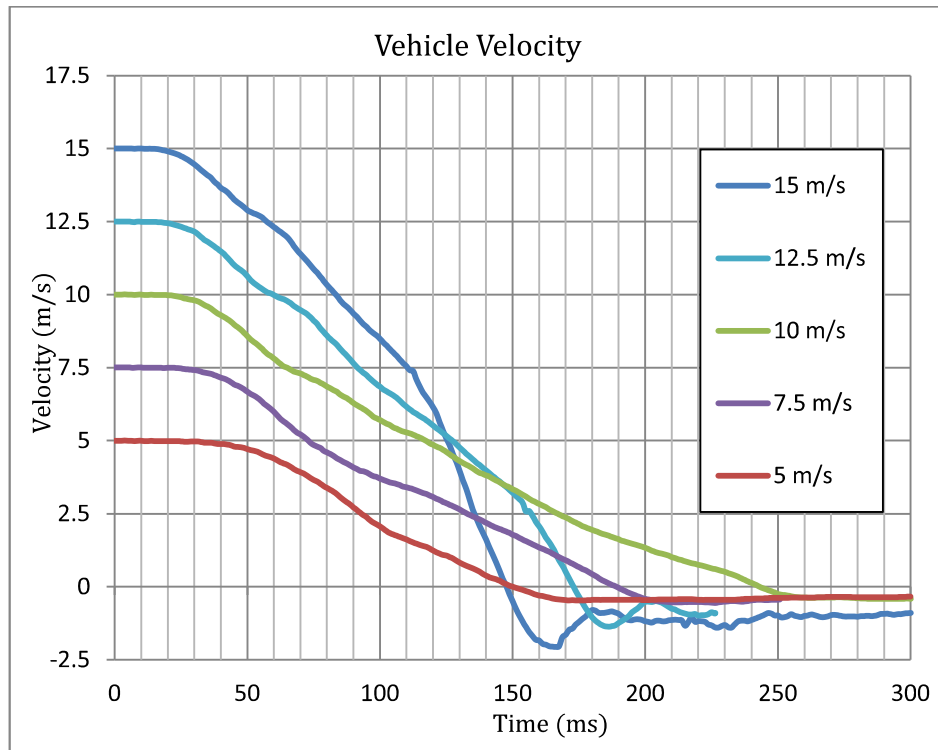


Figure 42: Velocity profile of the car during the impact.

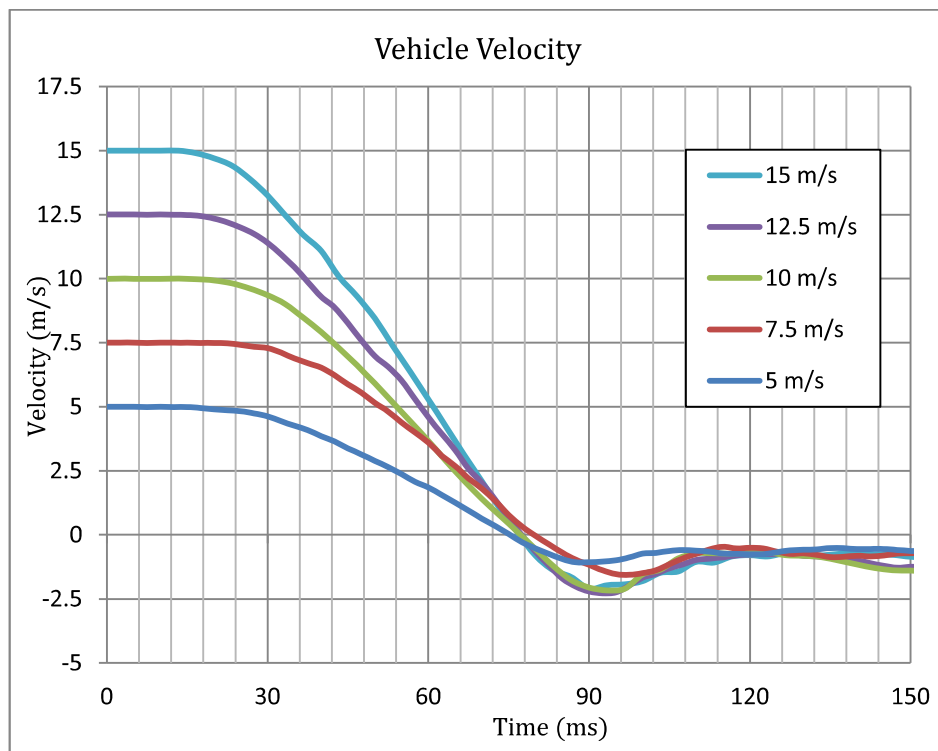


Figure 43: Velocity profile of the car during the impact with rigid wall.

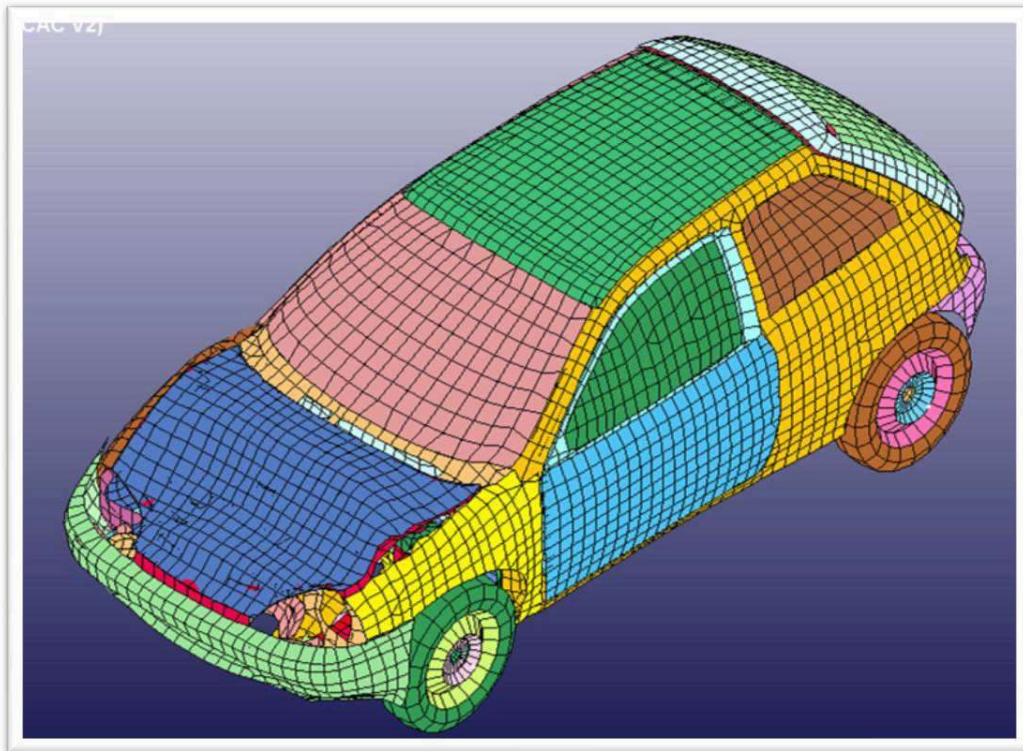


Figure 44: Final deformation of the vehicle after crashing the foam at 10 m/s.

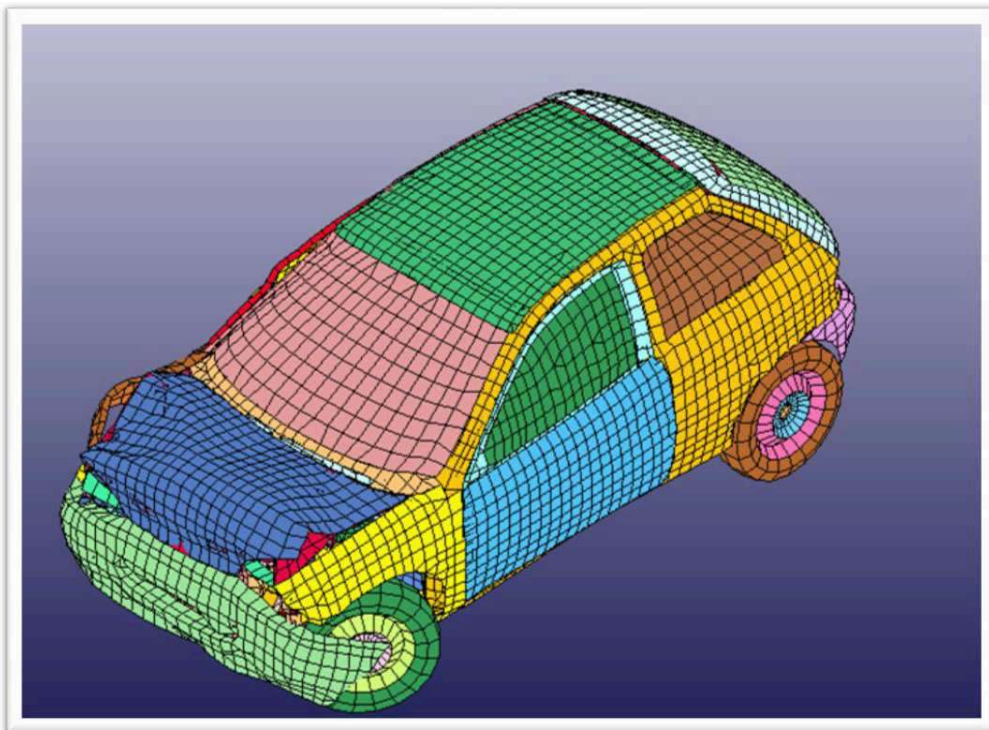


Figure 45: Final deformation of the vehicle after crashing the rigid wall directly at 10 m/s.

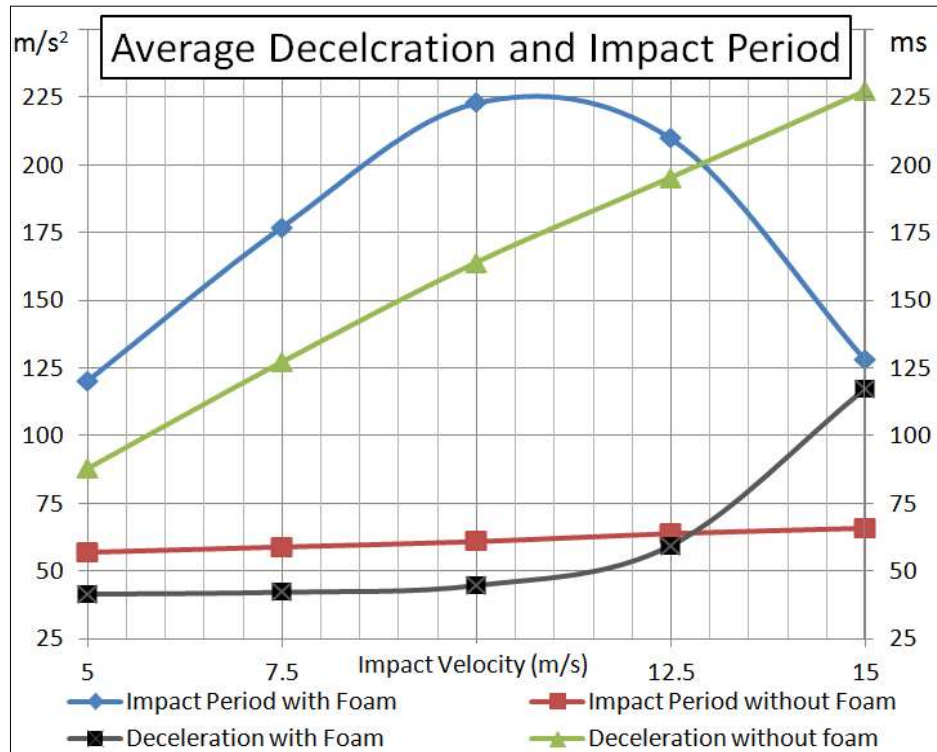


Figure 46: The effect of impact velocity on average declaration and impact period.

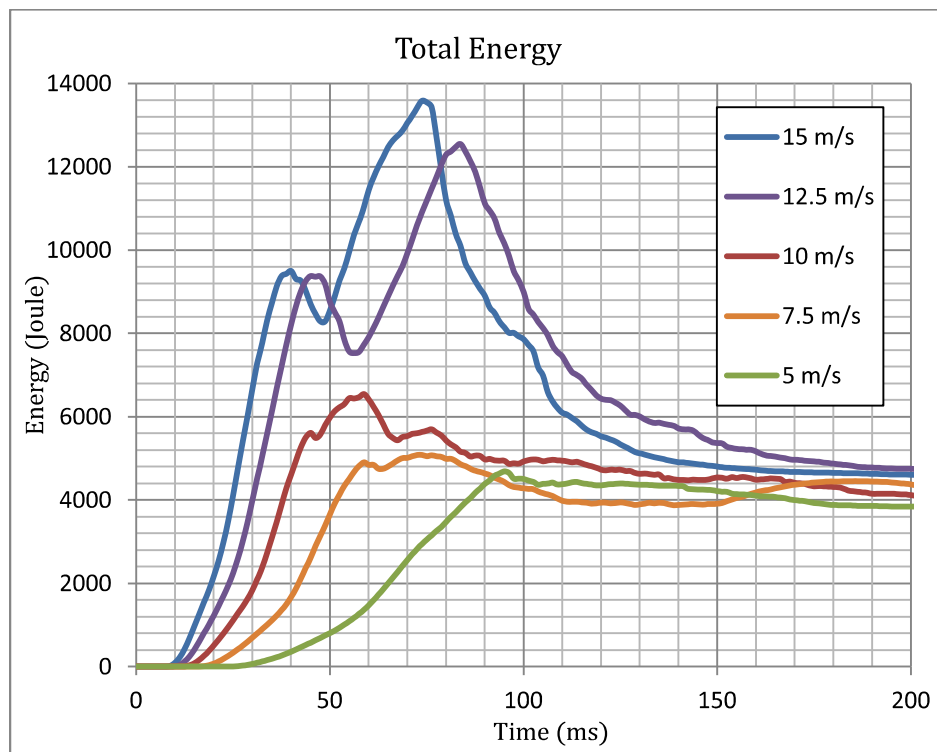


Figure 47: Total energy absorbed by the foam after impact.

CHAPTER 6

CONCLUSIONS

6.1 Crushable Foams

The crushable foams are polymeric materials which are synthesized by combining two or more materials. The mechanical behavior and the properties of these materials are designed and altered by engineers and researchers to achieve certain objectives. These types of materials have a low cost and weight, however a high energy absorption capability. As a result, they are used in a wide range of applications.

From the results of the experimental work in this project, it was concluded that the energy absorption capability of the expanded polystyrene crushable foam is related to its strain rate. As the compressed or deformed surface area increases, the energy absorbed by the crushable foam increases. Also, the numerical simulations show that the absorbed energy is proportional to the strain rate. This is due the strain rate dependency of crushable foams.

6.2 Experimental Work

Simple compression test was done to obtain the properties of the crushable foam. Then a gravity-driven drop test was carried on in order to verify the developed material model of the expanded polystyrene crushable foam. Since the objective is met as the simulation shows only a very small error, no further experimental work was done in this paper. However, more interesting experimental works are to be done in future work.

6.3 Numerical Simulation

The results of the simulation were satisfactory as they were validated by the experimental results. However, the behavior of the impact can be improved to cover more complicated shapes and larger range of velocities. This would be the main target of future work.

6.4 Recommendations

The study of energy absorption capabilities of the crushable foams surely is a very wide topic. A lot of further work must be done to fully understand these types of materials as this whole project focuses only on one type of the crushable foams. This report is a continuation of several works done by others and itself needs a continuation. The future work would be more productive and interesting.

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APPENDICES

(i) Reduced Input of Drop Test Simulation File

```

$# LS-DYNA Keyword file created by LS-PrePost 4.0 (Beta) - 07Aug2012(15:30)
$# Created on Dec-18-2012 (12:56:40)
*KEYWORD
*TITLE
$# title
LS-DYNA keyword deck by LS-PrePost
*CONTROL_CONTACT
$#  slsfac      rwpnal      islchk      shlthk      penopt      thkchg      orien      enmass
    0.100000      0.000          1          0          0          0          1          1
$#  usrstr      usrfric      nsbcs      interm      xpene      ssthk      ecddt      tiedprj
    0          0          0          0  4.000000          0          0          0
$#  sfric      dfrc      edc      vfc      th      th_sf      pen_sf
    0.000      0.000      0.000      0.000      0.000      0.000      0.000
$#  ignore      frceng      skiprwg      outseg      spotstp      spotdel      spothin
    0          0          0          0          0          0      0.000
$#  isym      nserod      rwgaps      rwgdlth      rwksf      icov      swradf      ithoff
    0          0          0      0.000  1.000000          0      0.000          0
$#  shldg      pstiff      ithcnt      tdcnof      ftall      unused      shltrw
    0          0          0          0          0          0      0.000
*CONTROL_ENERGY
$#  hgen      rwen      slnten      rylen
    2          2          2          2
*CONTROL_TERMINATION
$#  endtim      endcyc      dtmin      endeng      endmas
  100.00000          0      0.000      0.000      0.000
*CONTROL_TIMESTEP
$#  dtinit      tssfacc      isdo      tslimt      dt2ms      lctm      erode      ms1st
    0.000  0.500000          0      0.000      0.000          0          0          0
$#  dt2msf      dt2mslc      imsc1
    0.000          0          0
*DATABASE_BINARY_D3PLOT
$#  dt      lcdt      beam      npltc      psetid
    0.250000          0          0          0          0
$#  ioopt
    0

```

*DATABASE_EXTENT_BINARY

\$#	neiph	neips	maxint	strflg	sigflg	epsflg	rltflg	engflg
	0	0	3	1	1	1	1	1
\$#	cmpflg	ieverp	beamip	dcomp	shge	stssz	n3thdt	ialemat
	0	0	0	1	1	1	2	1
\$#	nintsl	pkp_sen	sclp	unused	msscl	therm	intout	nodout
	0	0	1.000000	0	0	0STRESS	STRESS	
\$#	dttd	resplt						
	0	0						

*LOAD_BODY_Z

\$#	lcid	sf	lcidrr	xc	yc	zc	cid
	2	1.000000	0	0.000	0.000	0.000	0

*CONTACT_INTERIOR

\$#	psid
	1

*CONTACT_AUTOMATIC_NODES_TO_SURFACE_ID

\$#	cid							title
	2							2NODE_TO_SURFACE
\$#	ssid	msid	sstyp	mstyp	sboxid	mboxid	spr	mpr
	2	1	4	3	0	0	0	0
\$#	fs	fd	dc	vc	vdc	penchk	bt	dt
	0.300000	0.300000	0.000	0.000	0.000	0	0.0001	0.0000E+20
\$#	sfs	sfm	sst	mst	sfst	sfmt	fsf	vsf
	1.000000	1.000000	0.000	0.000	1.000000	1.000000	1.000000	1.000000
\$#	soft	sofscl	lcidab	maxpar	sbopt	depth	bsort	frcfrq
	1	0.100000	0	1.025000	2.000000	2	0	1
\$#	penmax	thkopt	shlthk	snlog	isym	i2d3d	sldthk	sldstf
	0.000	0	0	0	0	0	1.500000	0.002060

*SET_NODE_LIST_TITLE

FOAM

\$#	sid	da1	da2	da3	da4	solver		
	2	0.000	0.000	0.000	0.000	MECH		
\$#	nid1	nid2	nid3	nid4	nid5	nid6	nid7	nid8
	2913710	2913711	2913712	2913713	2913714	2913715	2913716	2913717
	2913718	2913719	2913720	2913721	2913722	2913723	2913724	2913725

*PART

\$# title

Missile

\$#	pid	secid	mid	eosid	hgid	grav	adpopt	tmid
	1	1	1	0	0	0	0	0


```

*SECTION_SOLID_TITLE
Solid
$#   secid   elform   aet
      1       1       0

*MAT_RIGID_TITLE
Aluminum
$#   mid      ro      e      pr      n      couple      m      alias
      1 2.7000E-6 70.000000 0.340000 0.000 0.000 0.000
$#   cmo      con1     con2
      0.000      0      0
$# lco or a1     a2      a3      v1      v2      v3
      0.000      0.000 0.000 0.000 0.000 0.000

*PART
$# title
FOAM
$#   pid      secid      mid      eosid      hgid      grav      adpopt      tmid
      2       1       2       0       1       0       0       0

*MAT_CRUSHABLE_FOAM_TITLE
Crushable Foam
$#   mid      ro      e      pr      lcid      tsc      damp
      2 1.2750E-8 0.002060 0.000 1 1.0000E-4 0.500000

*HOURLASS_TITLE
Hourglass
$#   hgid      ihq      qm      ibq      q1      q2      qb/vdc      qw
      1       5 0.150000 0 1.500000 0.060000 0.100000 0.100000

*MAT_ADD_EROSION
$#   mid      excl     mxpres     mneps     effeps     voleps     numfip     ncs
      2 7777.0000 7777.0000 7777.0000 7777.0000 7777.0000 1.000000 2.000000
$# mnpres     sigpl     sigvm     mxeps     epssh     sigth     impulse     failtm
-3.250E-5 7777.0000 7777.0000 0.125000 7777.0000 7777.0000 7777.0000 7777.0000

*RIGIDWALL_PLANAR
$#   nsid      nsidex     boxid     offset     birth     death     rwksf
      0       0       0 0.000 0.0001.0000E+20 1.000000
$#   xt      yt      zt      xh      yh      zh      fric      wvel
16.000000-12.000000-435.00000 16.000000-12.000000-217.27000 0.000 0.000

*INITIAL_VELOCITY_GENERATION
$#nsid/pid      styp      omega      vx      vy      vz      ivatn      icid
      1       2 0.000 0.000 0.000-10.450000 0 0
$#   xc      yc      zc      nx      ny      nz      phase      iridid
      0.000      0.000 0.000 0.000 0.000 0.000 0 0

```

```

*DEFINE_CURVE_TITLE
Stress Strain
$#      lcid      sidr      sfa      sfo      offa      offo      dattyp
          1          0 1.000000  1.000000      0.000      0.000          0
$#          a1          o1
          0.000          0.000
          0.100212      7.350000e-005
          0.200425      9.130000e-005
          0.300425      1.100000e-004
          0.400425      1.350000e-004
          0.500425      1.700000e-004
          0.600425      2.230000e-004
          0.700424      3.110000e-004
          0.850000      7.010000e-004
          0.900000      1.040000e-003
          1.000000      2.470000e-003

*DEFINE_CURVE_TITLE
Gravity
$#      lcid      sidr      sfa      sfo      offa      offo      dattyp
          2          0 1.000000  1.000000      0.000      0.000          0
$#          a1          o1
          0.000          0.009810
          100.000000      0.009810

*SET_PART_LIST_TITLE
FOAM_PART_SET
$#      sid      da1      da2      da3      da4      solver
          1      0.000  0.100000      0.000  2.000000MECH
$#      pid1      pid2      pid3      pid4      pid5      pid6      pid7      pid8
          2          0          0          0          0          0          0          0

*ELEMENT_SOLID
$#      eid      pid      n1      n2      n3      n4      n5      n6      n7      n8
 3045158          2 2913710 2913711 2913772 2913771 2917431 2917432 2917493 2917492
 3045159          2 2913711 2913712 2913773 2913772 2917432 2917433 2917494 2917493

*NODE
$#      nid      x      y      z      tc      rc
 2913710     -120.000000     -120.000000     -435.000000      0      0
 2913711     -116.000000     -120.000000     -435.000000      0      0
 2913712     -112.000000     -120.000000     -435.000000      0      0

*END

```

(ii) Raw data of stress strain compression test at 500 mm/min

1	1		0.48	305.9705	1.580479
Time	Force	Stroke	0.49	311.2952	1.624958
sec	N	mm	0.5	318.7656	1.669719
0	-0.9934107	0	0.51	324.8453	1.714771
0.01	-0.9934107	0.0001875	0.52	332.7926	1.76
0.02	-0.9934107	0.001802083	0.53	341.8128	1.805437
0.03	-0.9934107	0.005927083	0.54	347.1375	1.851187
0.04	-0.9934107	0.01313542	0.55	353.4953	1.897167
0.05	-0.9934107	0.02359375	0.56	361.8002	1.943354
0.06	-0.9934107	0.03728125	0.57	369.7475	1.989833
0.07	-0.9934107	0.05404167	0.58	377.059	2.036562
0.08	-0.9934107	0.07360417	0.59	382.4631	2.08351
0.09	-0.9934107	0.09569792	0.6	389.7747	2.130656
0.1	-0.9934107	0.1200417	0.61	396.2914	2.178094
0.11	-0.9139379	0.1463854	0.62	400.8214	2.225802
0.12	-0.8742015	0.1744479	0.63	406.8613	2.273677
0.13	23.76238	0.2040312	0.64	413.4178	2.321844
0.14	34.41175	0.2349271	0.65	416.398	2.370281
0.15	40.84905	0.2669479	0.66	423.9082	2.418927
0.16	46.84925	0.2999687	0.67	429.4316	2.467802
0.17	54.20049	0.3338333	0.68	435.2729	2.516948
0.18	61.51199	0.3684583	0.69	440.5181	2.566344
0.19	67.71088	0.4037396	0.7	447.1938	2.615906
0.2	78.04235	0.4396354	0.71	452.439	2.665792
0.21	84.7578	0.4760312	0.72	457.3266	2.715937
0.22	94.53297	0.5129583	0.73	461.0221	2.766271
0.23	103.7518	0.5502709	0.74	464.956	2.816833
0.24	111.6991	0.5880208	0.75	470.0422	2.867687
0.25	116.984	0.6261563	0.76	475.4066	2.918781
0.26	125.6068	0.6645938	0.77	481.3671	2.970062
0.27	135.0244	0.7033958	0.78	484.7447	3.021667
0.28	142.2167	0.7425208	0.79	488.4005	3.07349
0.29	151.3163	0.7818854	0.8	493.8841	3.1255
0.3	160.4557	0.8215729	0.81	498.414	3.177792
0.31	168.8401	0.8615729	0.82	503.9374	3.230354
0.32	177.2642	0.9018125	0.83	507.1958	3.283135
0.33	185.291	0.9423125	0.84	509.1826	3.336115
0.34	194.3906	0.9831354	0.85	515.6994	3.389396
0.35	203.8479	1.024187	0.86	518.6002	3.442937
0.36	210.2057	1.065437	0.87	520.7062	3.496656
0.37	217.994	1.10699	0.88	526.1104	3.550646
0.38	225.4248	1.148802	0.89	529.17	3.604906
0.39	234.6436	1.190812	0.9	532.4681	3.659385
0.4	243.2664	1.233135	0.91	535.806	3.714094
0.41	252.2071	1.27574	0.92	539.4221	3.769094
0.42	259.3199	1.318542	0.93	542.8394	3.824302
0.43	264.5652	1.361542	0.94	547.0515	3.879719
0.44	271.8766	1.404854	0.95	552.5748	3.935427
0.45	281.771	1.448406	0.96	556.31	3.991396
0.46	290.3144	1.492156	0.97	556.5484	4.047563
0.47	296.6324	1.536188	0.98	558.8929	4.103958

0.99	563.701	4.160656	1.53	682.5924	7.581666
1	568.3104	4.217583	1.54	684.4203	7.651729
1.01	571.0125	4.274698	1.55	685.4534	7.722031
1.02	574.1119	4.332115	1.56	688.8707	7.792511
1.03	573.9133	4.389781	1.57	688.5926	7.863292
1.04	578.9598	4.447635	1.58	688.8707	7.934312
1.05	582.2181	4.50575	1.59	691.7715	8.005542
1.06	584.4037	4.564146	1.6	693.5199	8.077042
1.07	585.9534	4.62276	1.61	695.467	8.148802
1.08	586.9468	4.681583	1.62	697.0565	8.220781
1.09	589.1721	4.740708	1.63	699.2022	8.292958
1.1	591.0794	4.800073	1.64	699.9175	8.365458
1.11	595.4106	4.859604	1.65	703.454	8.438188
1.12	597.8743	4.919437	1.66	704.4871	8.511104
1.13	599.5035	4.979552	1.67	703.4938	8.584302
1.14	601.8877	5.039865	1.68	705.5203	8.657771
1.15	605.9408	5.100385	1.69	709.8119	8.731448
1.16	607.0535	5.161208	1.7	711.1232	8.805354
1.17	609.5568	5.222271	1.71	712.1165	8.879531
1.18	612.3781	5.283521	1.72	713.9445	8.953959
1.19	613.9676	5.345062	1.73	716.2889	9.028573
1.2	618.2592	5.406865	1.74	714.8981	9.10349
1.21	619.1333	5.468864	1.75	717.0042	9.178646
1.22	620.5241	5.531094	1.76	721.1367	9.254021
1.23	622.4711	5.593625	1.77	723.8388	9.329625
1.24	626.4448	5.656385	1.78	724.1964	9.40552
1.25	628.2727	5.719344	1.79	724.3951	9.481646
1.26	629.5045	5.782583	1.8	726.7396	9.557948
1.27	631.9285	5.846063	1.81	729.0443	9.634562
1.28	635.0279	5.909739	1.82	731.1106	9.711448
1.29	636.7762	5.973708	1.83	730.8324	9.7885
1.3	638.3657	6.037958	1.84	732.9781	9.865812
1.31	639.7168	6.102396	1.85	735.4021	9.943406
1.32	639.4386	6.167042	1.86	735.3624	10.02123
1.33	643.3328	6.232	1.87	739.6142	10.09924
1.34	645.9554	6.297198	1.88	741.0447	10.17756
1.35	648.4588	6.362583	1.89	742.8726	10.25613
1.36	648.1409	6.428239	1.9	744.0249	10.33488
1.37	652.79	6.494187	1.91	743.2699	10.41391
1.38	654.3795	6.560312	1.92	746.9654	10.49324
1.39	656.8432	6.626656	1.93	748.7933	10.57272
1.4	657.8763	6.693344	1.94	751.694	10.65246
1.41	659.6644	6.76024	1.95	752.3696	10.7325
1.42	662.0486	6.827312	1.96	753.6809	10.81275
1.43	664.5521	6.894677	1.97	754.9525	10.89318
1.44	665.466	6.962323	1.98	755.8267	10.97394
1.45	668.327	7.030156	1.99	755.9459	11.05495
1.46	669.3999	7.098218	2	758.7274	11.13612
1.47	670.2344	7.166573	2.01	760.1182	11.21757
1.48	673.0954	7.235177	2.02	763.6945	11.2993
1.49	674.6053	7.303948	2.03	765.5621	11.38125
1.5	675.5988	7.373031	2.04	766.1183	11.46336
1.51	679.4135	7.442375	2.05	766.5952	11.54558
1.52	682.2347	7.511896	2.06	769.2576	11.62802

2.07	772.2775	11.71072	2.61	852.6246	16.20913
2.08	773.2312	11.79358	2.62	853.1014	16.29246
2.09	775.1385	11.87656	2.63	852.3066	16.37578
2.1	775.9333	11.95967	2.64	853.0616	16.45911
2.11	777.8804	12.04284	2.65	854.5717	16.54246
2.12	779.8274	12.12605	2.66	854.2935	16.62577
2.13	781.5759	12.2093	2.67	855.2869	16.7091
2.14	781.7745	12.29256	2.68	858.5453	16.79245
2.15	782.6487	12.37586	2.69	859.1811	16.87576
2.16	786.7416	12.45918	2.7	860.2142	16.9591
2.17	787.4966	12.54249	2.71	865.0223	17.04246
2.18	788.0132	12.6258	2.72	866.0555	17.12578
2.19	790.5165	12.70915	2.73	864.2673	17.20911
2.2	791.8279	12.79248	2.74	864.1879	17.29246
2.21	792.6623	12.8758	2.75	866.9297	17.37578
2.22	792.5828	12.95914	2.76	869.1947	17.45913
2.23	794.7683	13.04248	2.77	871.1815	17.54245
2.24	796.9141	13.12579	2.78	872.4531	17.62577
2.25	798.0665	13.20912	2.79	873.4465	17.70913
2.26	798.8215	13.29246	2.8	873.367	17.79245
2.27	801.8812	13.37578	2.81	876.1088	17.87576
2.28	801.3646	13.45912	2.82	877.6586	17.95909
2.29	802.2785	13.54246	2.83	879.8043	18.04246
2.3	804.7025	13.62578	2.84	882.1885	18.12578
2.31	807.2853	13.70914	2.85	882.8242	18.2091
2.32	806.7687	13.79247	2.86	883.0627	18.29244
2.33	807.6827	13.87578	2.87	885.2482	18.37577
2.34	811.1000	13.95911	2.88	887.5529	18.4591
2.35	811.8152	14.04247	2.89	887.4734	18.54244
2.36	813.4842	14.12579	2.9	888.149	18.62576
2.37	814.5968	14.20911	2.91	888.2682	18.70911
2.38	816.4644	14.29245	2.92	889.1027	18.79245
2.39	818.2128	14.37577	2.93	891.765	18.87577
2.4	819.7228	14.45911	2.94	892.6789	18.9591
2.41	822.5839	14.54245	2.95	896.2155	19.04246
2.42	822.2659	14.62577	2.96	899.4738	19.12577
2.43	822.5441	14.70912	2.97	899.2354	19.20911
2.44	824.6501	14.79247	2.98	899.9904	19.29245
2.45	827.4714	14.87577	2.99	901.9375	19.37577
2.46	828.3854	14.95911	3.00	902.017	19.45911
2.47	830.8490	15.04246	3.01	902.0964	19.54245
2.48	832.7166	15.12578	3.02	903.2885	19.62577
2.49	832.3590	15.20911	3.03	908.0966	19.70913
2.50	832.8755	15.29245	3.04	909.7258	19.79246
2.51	834.028	15.37578	3.05	909.8848	19.87577
2.52	836.6902	15.45912	3.06	910.6795	19.9591
2.53	838.6771	15.54246	3.07	912.7458	20.04246
2.54	838.7566	15.62578	3.08	914.5737	20.12578
2.55	840.3063	15.70912	3.09	915.1697	20.20911
2.56	843.1276	15.79246	3.1	915.3684	20.29244
2.57	844.7568	15.87577	3.11	916.7989	20.37577
2.58	846.6244	15.95911	3.12	919.342	20.4591
2.59	847.2999	16.04246	3.13	923.5938	20.54244
2.6	850.2801	16.12578	3.14	925.6204	20.62576

3.15	924.0707	20.7091	3.69	1002.471	25.2091
3.16	927.2893	20.79244	3.7	1003.742	25.29244
3.17	927.6867	20.87576	3.71	1006.087	25.37577
3.18	927.8059	20.95909	3.72	1007.517	25.45911
3.19	926.6536	21.04245	3.73	1011.133	25.54245
3.2	929.5941	21.12577	3.74	1012.564	25.62576
3.21	931.7398	21.20911	3.75	1012.484	25.70911
3.22	932.4551	21.29245	3.76	1011.411	25.79246
3.23	934.9584	21.37577	3.77	1015.147	25.87578
3.24	935.9916	21.45911	3.78	1016.18	25.95911
3.25	937.6605	21.54245	3.79	1018.643	26.04245
3.26	939.4884	21.62577	3.8	1018.524	26.12577
3.27	939.3295	21.70913	3.81	1020.273	26.20911
3.28	942.27	21.79246	3.82	1024.365	26.29245
3.29	945.4886	21.87578	3.83	1027.346	26.37578
3.3	946.7602	21.95911	3.84	1026.829	26.45911
3.31	948.4689	22.04246	3.85	1026.551	26.54245
3.32	949.8596	22.12578	3.86	1027.068	26.62577
3.33	950.4557	22.20911	3.87	1029.571	26.70911
3.34	950.8928	22.29245	3.88	1033.107	26.79245
3.35	953.2372	22.37577	3.89	1034.617	26.87577
3.36	955.1843	22.45911	3.9	1035.81	26.95911
3.37	956.3367	22.54245	3.91	1037.677	27.04245
3.38	955.0651	22.62576	3.92	1039.426	27.12577
3.39	956.4161	22.70911	3.93	1039.982	27.20911
3.4	958.5222	22.79246	3.94	1042.286	27.29245
3.41	960.1514	22.87577	3.95	1043.757	27.37578
3.42	963.1713	22.95911	3.96	1044.83	27.45911
3.43	964.1647	23.04246	3.97	1046.101	27.54245
3.44	965.8337	23.12578	3.98	1049.399	27.62576
3.45	967.1052	23.20913	3.99	1049.081	27.70913
3.46	967.5424	23.29245	4	1051.426	27.79245
3.47	971.3173	23.37577	4.01	1053.572	27.87576
3.48	973.5823	23.45911	4.02	1054.128	27.9591
3.49	976.1651	23.54244	4.03	1058.221	28.04246
3.5	974.6154	23.62577	4.04	1059.771	28.12578
3.51	976.0062	23.70913	4.05	1059.929	28.2091
3.52	977.0393	23.79245	4.06	1060.923	28.29244
3.53	977.834	23.87577	4.07	1063.069	28.37576
3.54	978.1122	23.95911	4.08	1065.214	28.4591
3.55	981.6091	24.04246	4.09	1068.473	28.54244
3.56	984.9866	24.12578	4.1	1070.897	28.62576
3.57	984.8674	24.2091	4.11	1071.89	28.70911
3.58	987.9271	24.29244	4.12	1073.321	28.79246
3.59	988.6027	24.37578	4.13	1075.109	28.87576
3.6	988.0066	24.45911	4.14	1075.149	28.9591
3.61	989.2781	24.54245	4.15	1076.659	29.04246
3.62	992.8942	24.62577	4.16	1079.957	29.12577
3.63	993.967	24.70911	4.17	1081.943	29.2091
3.64	995.7552	24.79246	4.18	1080.314	29.29244
3.65	996.2321	24.87577	4.19	1082.659	29.37577
3.66	997.5036	24.9591	4.2	1087.546	29.4591
3.67	1001.04	25.04245	4.21	1089.851	29.54244
3.68	1000.404	25.12577	4.22	1092.196	29.62577

4.23	1092.593	29.70911	4.77	1192.808	34.2091
4.24	1091.242	29.79245	4.78	1192.967	34.29245
4.25	1093.467	29.87577	4.79	1195.312	34.37578
4.26	1096.686	29.95911	4.8	1199.404	34.4591
4.27	1098.792	30.04246	4.81	1200.676	34.54244
4.28	1099.269	30.12578	4.82	1202.901	34.62576
4.29	1102.408	30.20911	4.83	1205.603	34.7091
4.3	1104.434	30.29245	4.84	1208.583	34.79244
4.31	1106.62	30.37578	4.85	1210.491	34.87575
4.32	1107.613	30.45913	4.86	1211.564	34.9591
4.33	1108.487	30.54245	4.87	1213.352	35.04245
4.34	1110.355	30.62576	4.88	1214.782	35.12578
4.35	1111.388	30.70911	4.89	1218.557	35.20911
4.36	1114.408	30.79245	4.9	1220.862	35.29244
4.37	1116.435	30.87577	4.91	1221.339	35.37578
4.38	1116.832	30.9591	4.92	1223.365	35.45911
4.39	1119.296	31.04245	4.93	1225.352	35.54246
4.4	1120.726	31.12577	4.94	1229.445	35.62577
4.41	1122.514	31.2091	4.95	1230.439	35.70911
4.42	1124.104	31.29244	4.96	1231.71	35.79246
4.43	1124.938	31.37577	4.97	1235.763	35.87577
4.44	1129.588	31.45911	4.98	1239.101	35.95911
4.45	1131.614	31.54245	4.99	1240.492	36.04247
4.46	1132.329	31.62576	5	1242.081	36.12578
4.47	1133.839	31.70911	5.01	1241.406	36.20911
4.48	1133.045	31.79246	5.02	1243.711	36.29244
4.49	1137.217	31.87577	5.03	1247.446	36.37577
4.5	1138.727	31.9591	5.04	1251.459	36.4591
4.51	1143.535	32.04245	5.05	1251.578	36.54245
4.52	1146.356	32.12577	5.06	1253.366	36.62576
4.53	1146.754	32.2091	5.07	1259.247	36.70913
4.54	1148.78	32.29244	5.08	1259.406	36.79246
4.55	1151.363	32.37577	5.09	1260.718	36.87578
4.56	1151.443	32.4591	5.1	1263.539	36.95911
4.57	1152.953	32.54245	5.11	1266.996	37.04245
4.58	1155.376	32.62577	5.12	1268.903	37.12578
4.59	1159.191	32.70911	5.13	1270.692	37.20911
4.6	1161.178	32.79245	5.14	1273.076	37.29244
4.61	1161.059	32.87577	5.15	1276.334	37.37577
4.62	1160.502	32.9591	5.16	1277.328	37.45911
4.63	1162.847	33.04245	5.17	1278.241	37.54245
4.64	1167.099	33.12577	5.18	1280.665	37.62576
4.65	1170	33.20911	5.19	1283.765	37.70911
4.66	1171.986	33.29245	5.2	1286.745	37.79245
4.67	1174.966	33.37577	5.21	1287.778	37.87576
4.68	1174.847	33.4591	5.22	1289.606	37.9591
4.69	1176.198	33.54244	5.23	1290.798	38.04246
4.7	1179.496	33.62576	5.24	1293.739	38.12578
4.71	1181.046	33.70911	5.25	1296.56	38.20911
4.72	1182	33.79246	5.26	1298.308	38.29244
4.73	1184.622	33.87577	5.27	1300.534	38.37577
4.74	1188	33.9591	5.28	1304.15	38.45911
4.75	1190.98	34.04246	5.29	1305.66	38.54245
4.76	1192.053	34.12577	5.3	1306.733	38.62577

5.31	1310.587	38.70911	5.85	1446.962	43.20911
5.32	1312.693	38.79245	5.86	1449.148	43.29245
5.33	1314.918	38.87577	5.87	1452.605	43.37577
5.34	1318.773	38.9591	5.88	1455.466	43.45911
5.35	1320.759	39.04245	5.89	1457.969	43.54246
5.36	1322.23	39.12578	5.9	1459.917	43.62578
5.37	1324.932	39.2091	5.91	1465.479	43.70913
5.38	1326.919	39.29244	5.92	1465.996	43.79246
5.39	1330.217	39.37576	5.93	1467.029	43.87577
5.4	1331.846	39.4591	5.94	1469.89	43.95911
5.41	1333.912	39.54244	5.95	1472.592	44.04245
5.42	1336.535	39.62577	5.96	1475.533	44.12577
5.43	1338.085	39.70911	5.97	1479.705	44.2091
5.44	1341.78	39.79246	5.98	1484.315	44.29245
5.45	1345.038	39.87577	5.99	1485.626	44.37578
5.46	1347.621	39.9591	6	1488.487	44.45911
5.47	1347.82	40.04246	6.01	1492.421	44.54245
5.48	1351.237	40.12578	6.02	1495.6	44.62577
5.49	1353.184	40.2091	6.03	1496.991	44.70911
5.5	1356.443	40.29244	6.04	1498.342	44.79245
5.51	1359.9	40.37577	6.05	1502.792	44.87576
5.52	1361.211	40.4591	6.06	1506.368	44.9591
5.53	1363.158	40.54244	6.07	1507.998	45.04245
5.54	1365.185	40.62574	6.08	1510.223	45.12578
5.55	1367.569	40.7091	6.09	1514.594	45.20911
5.56	1371.543	40.79244	6.1	1517.892	45.29244
5.57	1373.251	40.87577	6.11	1520.236	45.37578
5.58	1376.192	40.9591	6.12	1522.66	45.45911
5.59	1377.424	41.04245	6.13	1525.005	45.54245
5.6	1379.927	41.12578	6.14	1528.104	45.62576
5.61	1385.45	41.20911	6.15	1532.594	45.70911
5.62	1387.477	41.29245	6.16	1535.416	45.79244
5.63	1388.669	41.37578	6.17	1536.846	45.87576
5.64	1393.437	41.45911	6.18	1538.912	45.9591
5.65	1395.146	41.54246	6.19	1543.244	46.04246
5.66	1396.656	41.62576	6.2	1545.27	46.12579
5.67	1399.716	41.70913	6.21	1549.363	46.20911
5.68	1401.186	41.79246	6.22	1552.86	46.29244
5.69	1404.087	41.87577	6.23	1556.834	46.37577
5.7	1407.504	41.9591	6.24	1557.191	46.4591
5.71	1409.849	42.04244	6.25	1560.569	46.54245
5.72	1413.703	42.12576	6.26	1564.622	46.62577
5.73	1417.081	42.2091	6.27	1565.854	46.70911
5.74	1416.644	42.29244	6.28	1570.9	46.79245
5.75	1418.829	42.37577	6.29	1576.344	46.87576
5.76	1422.366	42.45911	6.3	1577.338	46.9591
5.77	1426.101	42.54244	6.31	1581.43	47.04245
5.78	1429.836	42.62575	6.32	1584.808	47.12577
5.79	1432.101	42.7091	6.33	1587.152	47.2091
5.8	1434.803	42.79244	6.34	1589.179	47.29245
5.81	1436.472	42.87576	6.35	1593.947	47.37577
5.82	1438.26	42.9591	6.36	1597.444	47.45911
5.83	1442.393	43.04246	6.37	1598.557	47.54245
5.84	1446.168	43.12578	6.38	1599.908	47.62576

6.39	1603.405	47.70911	6.93	1793.861	52.20911
6.4	1609.484	47.79245	6.94	1798.232	52.29244
6.41	1612.981	47.87577	6.95	1801.252	52.37578
6.42	1616.279	47.95911	6.96	1806.935	52.45911
6.43	1619.101	48.04246	6.97	1811.862	52.54245
6.44	1620.809	48.12578	6.98	1815.557	52.62576
6.45	1624.028	48.20911	6.99	1818.538	52.7091
6.46	1628.161	48.29245	7	1824.141	52.79245
6.47	1631.22	48.37577	7.01	1828.313	52.87576
6.48	1634.161	48.4591	7.02	1831.651	52.9591
6.49	1638.929	48.54244	7.03	1835.028	53.04245
6.5	1643.538	48.62575	7.04	1837.651	53.12577
6.51	1645.406	48.7091	7.05	1841.784	53.2091
6.52	1649.38	48.79244	7.06	1845.479	53.29244
6.53	1652.837	48.87576	7.07	1850.247	53.37577
6.54	1656.135	48.9591	7.08	1854.579	53.45911
6.55	1657.605	49.04245	7.09	1857.877	53.54244
6.56	1661.062	49.12576	7.1	1863.003	53.62576
6.57	1665.314	49.2091	7.11	1867.572	53.70911
6.58	1669.288	49.29244	7.12	1872.46	53.79245
6.59	1673.778	49.37577	7.13	1876.831	53.87576
6.6	1678.109	49.45911	7.14	1881.838	53.95911
6.61	1678.745	49.54245	7.15	1884.143	54.04246
6.62	1680.414	49.62577	7.16	1888.593	54.12579
6.63	1685.46	49.70911	7.17	1893.361	54.2091
6.64	1690.547	49.79246	7.18	1896.183	54.29245
6.65	1692.732	49.87577	7.19	1900.395	54.37577
6.66	1695.236	49.9591	7.2	1903.931	54.4591
6.67	1698.534	50.04246	7.21	1909.296	54.54244
6.68	1702.786	50.12578	7.22	1913.23	54.62576
6.69	1705.964	50.2091	7.23	1918.157	54.7091
6.7	1710.971	50.29244	7.24	1921.455	54.79244
6.71	1716.534	50.37577	7.25	1928.608	54.87575
6.72	1717.011	50.4591	7.26	1932.224	54.9591
6.73	1718.76	50.54244	7.27	1936.436	55.04245
6.74	1723.806	50.62577	7.28	1941.601	55.12578
6.75	1729.17	50.70911	7.29	1943.827	55.20911
6.76	1733.502	50.79246	7.3	1948.913	55.29245
6.77	1735.926	50.87578	7.31	1955.867	55.37578
6.78	1739.144	50.9591	7.32	1956.979	55.45911
6.79	1742.8	51.04245	7.33	1959.682	55.54245
6.8	1746.178	51.12577	7.34	1967.748	55.62576
6.81	1749.674	51.2091	7.35	1971.086	55.70913
6.82	1754.284	51.29244	7.36	1974.225	55.79246
6.83	1757.026	51.37577	7.37	1979.629	55.87578
6.84	1759.211	51.45911	7.38	1983.841	55.95913
6.85	1762.867	51.54245	7.39	1988.729	56.04247
6.86	1768.192	51.62576	7.4	1993.219	56.12578
6.87	1771.768	51.70911	7.41	1996.04	56.20911
6.88	1775.861	51.79245	7.42	2001.365	56.29245
6.89	1780.709	51.87578	7.43	2007.882	56.37578
6.9	1783.292	51.95911	7.44	2012.451	56.45911
6.91	1786.709	52.04246	7.45	2016.664	56.54245
6.92	1789.848	52.12577	7.46	2018.412	56.62576

7.47	2022.584	56.70911	8.01	2311.031	61.20911
7.48	2028.108	56.79245	8.02	2315.958	61.29245
7.49	2034.664	56.87576	8.03	2320.329	61.37578
7.5	2039.909	56.9591	8.04	2326.171	61.45911
7.51	2043.724	57.04245	8.05	2332.647	61.54245
7.52	2048.254	57.12577	8.06	2338.648	61.62575
7.53	2053.499	57.2091	8.07	2344.171	61.7091
7.54	2058.904	57.29244	8.08	2348.94	61.79245
7.55	2063.553	57.37576	8.09	2355.258	61.87577
7.56	2068.122	57.45911	8.1	2362.132	61.9591
7.57	2073.288	57.54244	8.11	2369.125	62.04246
7.58	2079.368	57.62576	8.12	2375.205	62.12577
7.59	2083.818	57.70911	8.13	2381.206	62.2091
7.6	2086.123	57.79246	8.14	2387.166	62.29245
7.61	2090.533	57.87579	8.15	2392.848	62.37577
7.62	2095.461	57.95911	8.16	2398.412	62.45911
7.63	2102.772	58.04247	8.17	2404.571	62.54244
7.64	2109.289	58.12577	8.18	2412.438	62.62576
7.65	2112.826	58.2091	8.19	2420.147	62.7091
7.66	2118.389	58.29244	8.2	2425.353	62.79244
7.67	2124.349	58.37577	8.21	2432.466	62.87576
7.68	2128.403	58.4591	8.22	2438.108	62.95909
7.69	2134.403	58.54243	8.23	2443.433	63.04245
7.7	2140.681	58.62575	8.24	2448.241	63.12576
7.71	2143.82	58.7091	8.25	2453.645	63.20909
7.72	2148.748	58.79245	8.26	2463.341	63.29244
7.73	2153.595	58.87577	8.27	2470.573	63.37577
7.74	2158.125	58.9591	8.28	2474.864	63.45911
7.75	2164.523	59.04244	8.29	2482.931	63.54245
7.76	2171.318	59.12577	8.3	2490.123	63.62576
7.77	2175.45	59.20911	8.31	2495.686	63.70913
7.78	2180.854	59.29244	8.32	2503.117	63.79246
7.79	2184.629	59.37577	8.33	2510.269	63.87577
7.8	2190.987	59.45911	8.34	2516.707	63.95911
7.81	2195.954	59.54245	8.35	2522.19	64.04246
7.82	2199.928	59.62576	8.36	2530.138	64.12578
7.83	2208.352	59.70911	8.37	2537.449	64.20911
7.84	2215.505	59.79246	8.38	2541.9	64.29246
7.85	2221.743	59.87578	8.39	2551.635	64.37579
7.86	2224.366	59.95911	8.4	2558.708	64.45912
7.87	2227.227	60.04246	8.41	2566.258	64.54245
7.88	2232.989	60.12578	8.42	2572.973	64.62577
7.89	2241.254	60.20911	8.43	2580.563	64.70911
7.9	2247.254	60.29244	8.44	2584.616	64.79245
7.91	2252.062	60.37577	8.45	2591.848	64.87577
7.92	2256.95	60.4591	8.46	2600.272	64.95911
7.93	2263.188	60.54244	8.47	2605.955	65.04246
7.94	2269.069	60.62577	8.48	2612.67	65.12578
7.95	2275.03	60.70911	8.49	2619.982	65.20911
7.96	2279.48	60.79246	8.5	2628.724	65.29245
7.97	2283.971	60.87577	8.51	2636.353	65.37577
7.98	2290.805	60.9591	8.52	2642.949	65.45911
7.99	2296.13	61.04245	8.53	2651.85	65.54243
8	2303.68	61.12578	8.54	2660.076	65.62575

8.55	2664.566	65.70911	9.09	3132.145	70.20911
8.56	2671.917	65.79245	9.1	3142.118	70.29245
8.57	2680.818	65.87577	9.11	3151.695	70.37577
8.58	2689.203	65.95911	9.12	3163.695	70.45911
8.59	2695.839	66.04245	9.13	3173.431	70.54245
8.6	2703.031	66.12578	9.14	3183.047	70.62577
8.61	2711.495	66.20911	9.15	3194.292	70.70911
8.62	2718.409	66.29243	9.16	3205.021	70.79245
8.63	2726.873	66.37576	9.17	3214.637	70.87576
8.64	2736.529	66.45909	9.18	3224.214	70.95909
8.65	2742.688	66.54243	9.19	3233.194	71.04243
8.66	2750.118	66.62576	9.2	3244.956	71.12576
8.67	2758.94	66.70911	9.21	3255.089	71.20911
8.68	2765.775	66.79245	9.22	3265.262	71.29243
8.69	2774.119	66.87577	9.23	3275.474	71.37577
8.7	2781.51	66.95909	9.24	3287.713	71.45911
8.71	2790.332	67.04245	9.25	3300.309	71.54243
8.72	2799.153	67.12577	9.26	3309.965	71.62575
8.73	2809.008	67.20911	9.27	3321.568	71.70911
8.74	2816.24	67.29243	9.28	3332.972	71.79246
8.75	2821.565	67.37577	9.29	3343.542	71.87577
8.76	2830.505	67.45911	9.3	3355.066	71.95911
8.77	2839.446	67.54243	9.31	3366.391	72.04246
8.78	2848.188	67.62576	9.32	3376.842	72.12578
8.79	2856.095	67.70911	9.33	3389.915	72.20911
8.8	2865.235	67.79246	9.34	3399.134	72.29243
8.81	2875.845	67.87577	9.35	3410.538	72.37577
8.82	2883.156	67.95911	9.36	3425.002	72.45911
8.83	2892.296	68.04247	9.37	3436.009	72.54243
8.84	2898.892	68.12579	9.38	3447.93	72.62576
8.85	2907.674	68.20911	9.39	3460.606	72.70911
8.86	2916.018	68.29243	9.4	3472.765	72.79245
8.87	2927.224	68.37577	9.41	3485.004	72.87577
8.88	2936.085	68.45911	9.42	3494.263	72.95911
8.89	2943.833	68.54243	9.43	3505.627	73.04245
8.9	2953.251	68.62577	9.44	3518.899	73.12578
8.91	2962.867	68.70911	9.45	3531.734	73.20911
8.92	2969.384	68.79245	9.46	3544.648	73.29243
8.93	2976.894	68.87577	9.47	3555.457	73.37577
8.94	2988.418	68.95911	9.48	3569.007	73.45911
8.95	2999.028	69.04245	9.49	3582.517	73.54243
8.96	3007.094	69.12577	9.5	3594.16	73.62576
8.97	3015.717	69.20911	9.51	3608.386	73.70911
8.98	3025.532	69.29245	9.52	3619.115	73.79245
8.99	3036.539	69.37578	9.53	3632.744	73.87576
9	3044.963	69.45911	9.54	3644.506	73.95911
9.01	3053.427	69.54245	9.55	3656.189	74.04247
9.02	3063.242	69.62576	9.56	3670.096	74.12578
9.03	3073.216	69.70911	9.57	3683.726	74.20911
9.04	3082.911	69.79245	9.58	3700.495	74.29245
9.05	3093.918	69.87577	9.59	3712.058	74.37577
9.06	3103.097	69.95911	9.6	3723.105	74.45911
9.07	3113.429	70.04245	9.61	3737.052	74.54243
9.08	3123.8	70.12577	9.62	3752.43	74.62575

9.63	3766.775	74.70911	10.17	4665.931	79.20909
9.64	3780.007	74.79245	10.18	4687.945	79.29243
9.65	3795.783	74.87576	10.19	4708.568	79.37568
9.66	3808.618	74.95911	10.2	4727.006	79.45687
9.67	3823.479	75.04245	10.21	4748.742	79.53265
9.68	3835.003	75.12577	10.22	4772.266	79.60147
9.69	3849.387	75.20911	10.23	4792.094	79.66272
9.7	3865.123	75.29243	10.24	4810.85	79.71638
9.71	3879.229	75.37577	10.25	4828.93	79.76278
9.72	3892.978	75.45911	10.26	4842.361	79.80251
9.73	3910.502	75.54245	10.27	4852.573	79.83625
9.74	3923.297	75.62576	10.28	4861.355	79.86471
9.75	3936.966	75.70911	10.29	4869.421	79.88858
9.76	3952.305	75.79246	10.3	4873.713	79.90852
9.77	3970.504	75.87577	10.31	4874.349	79.92509
9.78	3985.047	75.95911	10.32	4875.064	79.93882
9.79	3999.551	76.04245	10.33	4873.594	79.95017
9.8	4013.578	76.12578	10.34	4873.514	79.95954
9.81	4030.267	76.20911	10.35	4868.507	79.96725
9.82	4044.215	76.29245	10.36	4864.295	79.97356
9.83	4059.672	76.37578	10.37	4858.971	79.97877
9.84	4077.991	76.45911	10.38	4855.474	79.983
9.85	4094.402	76.54245	10.39	4848.52	79.98648
9.86	4109.859	76.62577	10.4	4841.646	79.98933
9.87	4126.032	76.70911	10.41	4837.116	79.99158
9.88	4141.768	76.79243	10.42	4830.162	79.99344
9.89	4156.351	76.87576	10.43	4823.009	79.995
9.9	4172.405	76.95911	10.44	4816.572	79.99635
9.91	4190.961	77.04245	10.45	4810.413	79.99741
9.92	4207.293	77.12577	10.46	4805.604	79.99817
9.93	4224.419	77.20911	10.47	4802.068	79.99871
9.94	4241.347	77.29243	10.48	4796.028	79.99909
9.95	4261.017	77.37577	10.49	4790.863	79.99937
9.96	4277.349	77.45911	10.5	4786.292	79.99955
9.97	4293.839	77.54243	10.51	4781.008	79.99969
9.98	4310.37	77.62577	10.52	4777.908	79.99978
9.99	4327.377	77.70912	10.53	4771.908	79.99984
10	4344.106	77.79247	10.54	4766.186	79.99989
10.01	4362.941	77.87579	10.55	4761.338	79.99992
10.02	4383.445	77.95911	10.56	4758.477	79.99995
10.03	4402.518	78.04246	10.57	4755.735	79.99997
10.04	4419.168	78.12577	10.58	4752.278	79.99997
10.05	4436.056	78.20911	10.59	4750.053	79.99999
10.06	4452.626	78.29243	10.6	4745.006	80
10.07	4472.891	78.37576	10.61	4739.205	80
10.08	4493.157	78.45911	10.62	4736.662	80
10.09	4511.277	78.54243	10.63	4731.297	80
10.1	4530.907	78.62576	10.64	4728.159	80
10.11	4550.537	78.70911			
10.12	4569.332	78.79246			
10.13	4587.134	78.87575			
10.14	4607.956	78.95909			
10.15	4627.585	79.04243			
10.16	4648.089	79.12576			